

2018 ESC Guidelines for the diagnosis and management of syncope *Free*

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The Task Force for the diagnosis and management of syncope of the European Society of Cardiology (ESC)

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The disclosure forms of all experts involved in the development of these Guidelines are available on the ESC website <http://www.escardio.org/guidelines>.

 For the Supplementary Data which include background information and detailed discussion of the data that have provided the basis for the Guidelines see <https://academic.oup.com/eurheartj/article-lookup/doi/10.1093/eurheartj/ehy037#supplementary-data>

 Click here to access the corresponding chapter in [ESC CardioMed - Section 39 Syncope](#)

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Abbreviations and acronyms

Abbreviations and Acronyms

Abbreviations and Acronyms

[A](#)

- ABPM

Ambulatory blood pressure monitoring

[Â](#)

- AF

Atrial fibrillation

[Â](#)

- ARVC

Arrhythmogenic right ventricular cardiomyopathy

[Â](#)

- ATP

Adenosine triphosphate

[Â](#)

- AV

Atrioventricular

[Â](#)

- AVID

Antiarrhythmics versus Implantable Defibrillators trial

[Â](#)

- BBB

Bundle branch block

[Â](#)

- BNP

B-type natriuretic peptide

[Â](#)

- BP

Blood pressure

[Â](#)

- b.p.m.

Beats per minute

[Â](#)

- CI

Confidence interval

[Â](#)

- CI-CSS

Cardioinhibitory carotid sinus syndrome

[Â](#)

- CPG

Committee for Practice Guidelines

[Â](#)

- CRT-D

Cardiac resynchronization therapy defibrillator

[Â](#)

- CSM

Carotid sinus massage

[Â](#)

- CSS

Carotid sinus syndrome

[Â](#)

- DCM

Dilated cardiomyopathy

[Â](#)

- DDD-PM

Dual chamber pacemaker

[Â](#)

- ECG

Electrocardiogram/electrocardiographic

[Â](#)

- ED

Emergency department

[Â](#)

- EEG

Electroencephalogram

[Â](#)

- EFAS

European Federation of Autonomic Societies

[Â](#)

- EFIM

European Federation of Internal Medicine

[Â](#)

- EHRA

European Heart Rhythm Association

[Â](#)

- ENS

European Neurological Society

[Â](#)

- EPS

Electrophysiological study

[Â](#)

- ESC

European Society of Cardiology

[Â](#)

- EUGMS

European Union Geriatric Medicine Society

[Â](#)

- EuSEM

European Society of Emergency Medicine

[Â](#)

- HBPM

Home blood pressure monitoring

[Â](#)

- HCM

Hypertrophic cardiomyopathy

[Â](#)

- HR

Heart rate

[Â](#)

- ICD

Implantable cardioverter defibrillator

[Â](#)

- ILR

Implantable loop recorder

[Â](#)

- ISSUE

International Study on Syncope of Unknown Etiology

[Â](#)

- L-DOPA

L-3,4-dihydroxyphenylalanine

[Â](#)

- LOC

Loss of consciousness

[Â](#)

- LQTS

Long QT syndrome

[Â](#)

- LVEF

Left ventricular ejection fraction

[Â](#)

- MRI

Magnetic resonance imaging

[Â](#)

- NYHA

New York Heart Association

[Â](#)

- OH

Orthostatic hypotension

[Â](#)

- PC-Trial

Physical Counterpressure Manoeuvres Trial

[Â](#)

- PCM

Physical counter-pressure manoeuvres

[Â](#)

- PNES

Psychogenic non-epileptic seizures

[Â](#)

- POST

Prevention of Syncope Trial

[Â](#)

- POTS

Postural orthostatic tachycardia syndrome

[Â](#)

- PPS

Psychogenic pseudosyncope

[Â](#)

- RCT

Randomized controlled trial

[Â](#)

- SCD

Sudden cardiac death

[Â](#)

- SNRT

Sinus node recovery time

[Â](#)

- SU

Syncope unit

[Â](#)

- SUP

Syncope Unit Project

[Â](#)

- SVT

Supraventricular tachycardia

[Â](#)

- TIA

Transient ischaemic attack

[Â](#)

- t.i.d.

Ter in die (three times daily)

[Â](#)

- TLOC

Transient loss of consciousness

[Â](#)

- TNG

Trinitroglycerin

[Â](#)

- VA

Ventricular arrhythmia

[Â](#)

- VF

Ventricular fibrillation

[Â](#)

- VT

Ventricular tachycardia

[Â](#)

- VVS

Vasovagal syncope

1. Preamble

Guidelines summarize and evaluate available evidence with the aim of assisting health professionals in selecting the best management strategies for an individual patient with a given condition. Guidelines and their recommendations should facilitate decision making of health professionals in their daily practice. However, the final decisions concerning an individual patient must be made by the responsible health professional(s) in consultation with the patient and caregiver as appropriate.

A great number of guidelines have been issued in recent years by the European Society of Cardiology (ESC), as well as by other societies and organisations. Because of the impact on clinical practice, quality criteria for the development of guidelines have been established in order to make all decisions transparent to the user. The recommendations for formulating and issuing ESC Guidelines can be found on the ESC website (<http://www.escardio.org/Guidelines-&-Education/Clinical-Practice-Guidelines/Guidelines-development/Writing-ESC-Guidelines>). ESC Guidelines represent the official position of the ESC on a given topic and are regularly updated.

Members of this Task Force were selected by the ESC, including representation from its relevant ESC sub-specialty groups, in order to represent professionals involved with the medical care of patients with this pathology. Selected experts in the field undertook a comprehensive review of the published evidence for management of a given condition according to ESC Committee for Practice Guidelines (CPG) policy. A critical evaluation of diagnostic and therapeutic procedures was performed, including assessment of the riskâbenefit ratio. The level of evidence and the strength of the recommendation of particular management options were weighed and graded according to predefined scales, as outlined in *Tables 1* and *2*.

Table 1

[Open in new tab](#)

Classes of recommendations

 graphic
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Table 1

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Classes of recommendations

 graphic
 graphic

Table 2

[Open in new tab](#)

Levels of evidence

 graphic
 graphic

Table 2

[Open in new tab](#)

Levels of evidence

 graphic



The experts of the writing and reviewing panels provided declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. These forms were compiled into one file and can be found on the ESC website (<http://www.escardio.org/guidelines>). Any changes in declarations of interest that arise during the writing period were notified to the ESC and updated. The Task Force received its entire financial support from the ESC without any involvement from the healthcare industry.

The ESC CPG supervises and coordinates the preparation of new Guidelines. The Committee is also responsible for the endorsement process of these Guidelines. The ESC Guidelines undergo extensive review by the CPG and external experts. After appropriate revisions the Guidelines are approved by all the experts involved in the Task Force. The finalized document is approved by the CPG for publication in the *European Heart Journal*. The Guidelines were developed after careful consideration of the scientific and medical knowledge and the evidence available at the time of their dating.

The task of developing ESC Guidelines also includes the creation of educational tools and implementation programmes for the recommendations including condensed pocket guideline versions, summary slides, booklets with essential messages, summary cards for non-specialists and an electronic version for digital applications (smartphones, etc.). These versions are abridged and thus, if needed, one should always refer to the full-text version, which is freely available via the ESC website and hosted on the EHJ website. The National Societies of the ESC are encouraged to endorse, translate and implement all ESC Guidelines. Implementation programmes are needed because it has been shown that the outcome of disease may be favourably influenced by the thorough application of clinical recommendations.

Surveys and registries are needed to verify that real-life daily practice is in keeping with what is recommended in the guidelines, thus completing the loop between clinical research, writing of guidelines, disseminating them and implementing them into clinical practice.

Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgment, as well as in the determination and the implementation of preventive, diagnostic, or therapeutic medical strategies. However, the ESC Guidelines do not override in any way whatsoever the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient or the patient's caregiver where appropriate and/or necessary. It is also the health professional's responsibility to verify the rules and regulations applicable to drugs and devices at the time of prescription.

2. Introduction

The first ESC Guidelines for the management of syncope were published in 2001, with subsequent versions in 2004 and 2009. In March 2015, the ESC CPG considered that there were enough new data to justify the production of new Guidelines.

The most important aspect characterizing this document is the composition of the Task Force, which is truly multidisciplinary. Cardiologists form a minority of the panel; experts in emergency medicine, internal medicine and physiology, neurology and autonomic diseases, geriatric medicine, and nursing cover all aspects of management of the various forms of syncope and transient loss of consciousness (TLOC).

Compared with the previous versions of these Guidelines, the 2018 document contains [Supplementary Data](#) as an integral part. While the print text mainly aims to give formal evidence-based recommendations according to the standardized rules of the ESC, this new web-only feature allows expansion of the content to practical issues, and aims to fill the gap between the best available scientific evidence and the need for dissemination of these concepts into clinical practice (*“We have the knowledge, we need to teach it”*). Thanks to the [Supplementary Data](#) further explanation on specific points is given, and thanks to the [Web Practical Instructions](#) advice is given on how to evaluate patients with loss of consciousness (LOC), and how to perform and interpret tests properly; whenever possible, we provide tracings, videos, flow charts, and checklists.

The document aims to be patient-orientated and focused on therapy, and to reduce the risk of recurrence and the life-threatening consequences of syncope recurrence. For this purpose, even in the absence of strong evidence from trials, we give as much advice as possible on the most appropriate therapy based on the practical expertise of the members of the Task Force (*“Our patients seek solutions, not only explanations”*). When possible, we provide therapeutic and decision-making algorithms.

Finally, we recognize that one major challenge in syncope management is the reduction of inappropriate admissions and inappropriate use of tests while maintaining the safety of the patient. We give strong focus to pathways and organizational issues (*“We have the knowledge; we need to apply it”*). In particular, we propose a care pathway for the management of patients with TLOC from their arrival in the emergency department (ED), and give practical instructions on how to set up outpatient syncope clinics (syncope units) aimed at reducing hospitalization, under- and misdiagnoses, and costs.

2.1 What is new in the 2018 version?

The changes in recommendations made in the 2018 version compared with the 2009 version, the new recommendations, and the most important new/revised concepts are summarized in [Figure 1](#).

 What is new in the 2018 syncope Guidelines? AA = antiarrhythmic; AF = atrial fibrillation; ARVC = arrhythmogenic right ventricular cardiomyopathy; CSM = carotid sinus massage; ECG = electrocardiogram; ED = emergency department; LVEF = left ventricular ejection fraction; EPS = electrophysiological study; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; OH = orthostatic hypotension; PCM = physical counter-pressure manoeuvres; POTS = postural orthostatic tachycardia syndrome; PPS = psychogenic pseudosyncope; SNRT = sinus node recovery time; SU = syncope unit; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

Figure 1

What is new in the 2018 syncope Guidelines? AA = antiarrhythmic; AF = atrial fibrillation; ARVC = arrhythmogenic right ventricular cardiomyopathy; CSM = carotid sinus massage; ECG = electrocardiogram; ED = emergency department; LVEF = left ventricular ejection fraction; EPS = electrophysiological study; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; OH = orthostatic hypotension; PCM = physical counter-pressure manoeuvres; POTS = postural orthostatic tachycardia syndrome; PPS = psychogenic pseudosyncope; SNRT = sinus node recovery time; SU = syncope unit; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

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 New/revised concepts in the management of syncope. ECG = electrocardiogram; ED = emergency department; ICD = implantable cardioverter defibrillator; SCD = sudden cardiac death.

Central illustration

New/revised concepts in the management of syncope. ECG = electrocardiogram; ED = emergency department; ICD = implantable cardioverter defibrillator; SCD = sudden cardiac death.

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3. Definitions, classification, and pathophysiology

3.1 Definitions

- *Syncope* is defined as TLOC due to cerebral hypoperfusion, characterized by a rapid onset, short duration, and spontaneous complete recovery.

Syncope shares many clinical features with other disorders; it therefore presents in many differential diagnoses. This group of disorders is labelled TLOC.

- *TLOC* is defined as a state of real or apparent LOC with loss of awareness, characterized by amnesia for the period of unconsciousness, abnormal motor control, loss of responsiveness, and a short duration.

The two main groups of TLOC are â€˜TLOC due to head traumaâ€™ and â€˜non-traumatic TLOCâ€™ (Figure 2). Traumatic TLOC will not be considered further in this document, so TLOC will be used to mean non-traumatic TLOC.

 Syncope in the context of transient loss of consciousness. Non-traumatic transient loss of consciousness is classified into one of four groupings: syncope, epileptic seizures, psychogenic transient loss of consciousness, and a miscellaneous group of rare causes. This order represents their rate of occurrence. Combinations occur; e.g. non-traumatic transient loss of consciousness causes can cause falls with concussion, in which case transient loss of consciousness is both traumatic and non-traumatic. TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

Figure 2

Syncope in the context of transient loss of consciousness. Non-traumatic transient loss of consciousness is classified into one of four groupings: syncope, epileptic seizures, psychogenic transient loss of consciousness, and a miscellaneous group of rare causes. This order represents their rate of occurrence. Combinations occur; e.g. non-traumatic transient loss of consciousness causes can cause falls with concussion, in which case transient loss of consciousness is both traumatic and non-traumatic. TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

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The clinical features characterizing TLOC are usually derived from history taking from patients and eyewitnesses. Specific characteristics that aid diagnosis are outlined in section 3 of the [Web Practical Instructions](#).

TLOC groups are defined using pathophysiology: the qualifying criterion for syncope is cerebral hypoperfusion; for epileptic seizures, it is abnormal excessive brain activity; and for psychogenic TLOC it is the psychological process of conversion. The syncope definition rests on pathophysiology because no set of clinical features encompasses all forms of syncope while also excluding all epileptic seizures and psychogenic TLOC events.

- The adjective *presyncope* is used to indicate symptoms and signs that occur before unconsciousness in syncope. Note that the noun *presyncope* is often used to describe a state that resembles the prodrome of syncope, but which is not followed by LOC.

A variety of terms are used that generally do not match the definitions in this document closely enough to be used as synonyms of the defined terms. For example, a â€˜faintâ€™ approximately conforms to syncope but emphasizes vasovagal syncope (VVS) over other forms. A glossary of uncertain terms is shown in section 1 of the [Web Practical Instructions](#).

3.2 Classification and pathophysiology of syncope and transient loss of consciousness

3.2.1 Syncope

Table 3 provides a classification of the principal causes of syncope, emphasizing groups of disorders with common pathophysiology, presentation, and risk. Clinical features, epidemiology, prognosis, impact on quality of life, and economic issues are shown in section 2 of the [Web Practical Instructions](#).

Table 3

[Open in new tab](#)

Classification of syncope

Reflex (neurally mediated) syncope Vasovagal: • orthostatic VVS: standing, less common sitting • emotional: fear, pain (somatic or visceral), instrumentation, blood phobia • Situational: ◦ micturition ◦ gastrointestinal stimulation (swallow, defaecation) ◦ cough, sneeze ◦ post-exercise ◦ others (e.g. laughing, brass instrument playing) • Carotid sinus syndrome • Non-classical forms (without prodromes and/or without apparent triggers and/or atypical presentation)
Syncope due to OH <i>Note that hypotension may be exacerbated by venous pooling during exercise (exercise-induced), after meals (postprandial hypotension), and after prolonged bed rest (deconditioning).</i> • Drug-induced OH (most common cause of OH): ◦ e.g. vasodilators, diuretics, phenothiazine, antidepressants • Volume depletion: ◦ haemorrhage, diarrhoea, vomiting, etc.

- Â â^{PADHTJ} Primary autonomic failure (neurogenic OH):
 - â^{PADNEH}â^{PADNEH} pure autonomic failure, multiple system atrophy, Parkinsonâ^{PADSGC}s disease, dementia with Lewy bodies
- Â â^{PADHTJ} Secondary autonomic failure (neurogenic OH):
 - â^{PADNEH}â^{PADNEH} diabetes, amyloidosis, spinal cord injuries, auto-immune autonomic neuropathy, paraneoplastic autonomic neuropathy, kidney failure

Cardiac syncope

Arrhythmia as primary cause:

- Bradycardia:
 - â^{PADNEH}â^{PADNEH} sinus node dysfunction (including bradycardia/tachycardia syndrome)
 - â^{PADNEH}â^{PADNEH} atrioventricular conduction system disease
- Â â^{PADHTJ} Tachycardia:
 - â^{PADNEH}â^{PADNEH} supraventricular
 - â^{PADNEH}â^{PADNEH} ventricular
- Â â^{PADHTJ} Structural cardiac: aortic stenosis, acute myocardial infarction/ischaemia, hypertrophic cardiomyopathy, cardiac masses (atrial myxoma, tumours, etc.), pericardial disease/tamponade, congenital anomalies of coronary arteries, prosthetic valve dysfunction
- Â â^{PADHTJ} Cardiopulmonary and great vessels: pulmonary embolus, acute aortic dissection, pulmonary hypertension

Remarks

- All forms of syncope, but mostly reflex syncope and OH, are more likely to occur or are more severe when various factors are present: medication causing low BP (due to vasodilatation or hypovolaemia), alcohol use, volume depletion (haemorrhage, low fluid intake, diarrhoea, vomiting), pulmonary diseases causing reduction in brain oxygen supply, environmental factors (thermal stress).
- There are two main pathophysiological mechanisms in reflex syncope. â^{PADST} Vasodepressionâ^{PADOSC} refers to conditions in which insufficient sympathetic vasoconstriction results in hypotension.¹ Â , Â ² â^{PADST} Cardioinhibitionâ^{PADOSC} is used when bradycardia or asystole predominates, reflecting a shift towards parasympathetic predominance. The haemodynamic pattern, i.e. cardioinhibitory,

vasodepressive, or both, is independent of the trigger evoking reflex syncope. For example, micturition syncope and orthostatic VVS may equally well present as cardioinhibitory or as vasodepressor syncope

- The non-classical form of reflex syncope involves a heterogeneous group of patients. The term is used to describe reflex syncope that occurs with uncertain or apparently absent triggers and/or atypical presentation. The diagnosis of reflex syncope is probable when other causes of syncope are excluded (absence of structural heart disease) and/or symptoms are reproduced in the tilt test.³ At present, this group also contains syncope associated with low adenosine plasma levels⁴ & ⁵
- The cardiovascular causes of orthostatic intolerance include classical OH, initial OH, delayed OH, POTS, and VVS, which in this context can be called orthostatic VVS.⁶ & ⁷ Syndromes of orthostatic intolerance that may cause syncope are presented in [Web Practical Instruction section 2](#).

Reflex (neurally mediated) syncope

Vasovagal:

- orthostatic VVS: standing, less common sitting
- emotional: fear, pain (somatic or visceral), instrumentation, blood phobia
- Situational:
 - o micturition
 - o gastrointestinal stimulation (swallow, defaecation)
 - o cough, sneeze
 - o post-exercise
 - o others (e.g. laughing, brass instrument playing)
 - o Carotid sinus syndrome
 - o Non-classical forms (without prodromes and/or without apparent triggers and/or atypical presentation)

Syncope due to OH

Note that hypotension may be exacerbated by venous pooling during exercise (exercise-induced), after meals (postprandial hypotension), and after prolonged bed rest (deconditioning).

- Drug-induced OH (most common cause of OH):

- ~~â~~~~â~~ e.g. vasodilators, diuretics, phenothiazine, antidepressants
- ~~Â~~~~â~~ Volume depletion:
 - ~~â~~~~â~~ haemorrhage, diarrhoea, vomiting, etc.
- ~~Â~~~~â~~ Primary autonomic failure (neurogenic OH):
 - ~~â~~~~â~~ pure autonomic failure, multiple system atrophy, Parkinson's disease, dementia with Lewy bodies
- ~~Â~~~~â~~ Secondary autonomic failure (neurogenic OH):
 - ~~â~~~~â~~ diabetes, amyloidosis, spinal cord injuries, auto-immune autonomic neuropathy, paraneoplastic autonomic neuropathy, kidney failure

Cardiac syncope

Arrhythmia as primary cause:

- Bradycardia:
 - ~~â~~~~â~~ sinus node dysfunction (including bradycardia/tachycardia syndrome)
 - ~~â~~~~â~~ atrioventricular conduction system disease
- ~~Â~~~~â~~ Tachycardia:
 - ~~â~~~~â~~ supraventricular
 - ~~â~~~~â~~ ventricular
- ~~Â~~~~â~~ Structural cardiac: aortic stenosis, acute myocardial infarction/ischaemia, hypertrophic cardiomyopathy, cardiac masses (atrial myxoma, tumours, etc.), pericardial disease/tamponade, congenital anomalies of coronary arteries, prosthetic valve dysfunction
- ~~Â~~~~â~~ Cardiopulmonary and great vessels: pulmonary embolus, acute aortic dissection, pulmonary hypertension

Remarks

- All forms of syncope, but mostly reflex syncope and OH, are more likely to occur or are more severe when various factors are present: medication causing low BP (due to vasodilatation or hypovolaemia), alcohol use, volume depletion (haemorrhage, low fluid intake, diarrhoea, vomiting), pulmonary diseases causing reduction in brain oxygen supply, environmental factors (thermal stress).

- There are two main pathophysiological mechanisms in reflex syncope. ¹ Vasodepression refers to conditions in which insufficient sympathetic vasoconstriction results in hypotension. ² Cardioinhibition is used when bradycardia or asystole predominates, reflecting a shift towards parasympathetic predominance. The haemodynamic pattern, i.e. cardioinhibitory, vasodepressive, or both, is independent of the trigger evoking reflex syncope. For example, micturition syncope and orthostatic VVS may equally well present as cardioinhibitory or as vasodepressor syncope
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BP = blood pressure; OH = orthostatic hypotension; POTS = postural orthostatic tachycardia syndrome; VVS = vasovagal syncope.

Table 3

[Open in new tab](#)

Classification of syncope

Reflex (neurally mediated) syncope

Vasovagal:

- orthostatic VVS: standing, less common sitting
- emotional: fear, pain (somatic or visceral), instrumentation, blood phobia
- Situational:
 - micturition
 - gastrointestinal stimulation (swallow, defaecation)
 - cough, sneeze
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 - others (e.g. laughing, brass instrument playing)

- Â â^{PADHTJ} Carotid sinus syndrome
- Â â^{PADHTJ} Non-classical forms (without prodromes and/or without apparent triggers and/or atypical presentation)

Syncope due to OH

Note that hypotension may be exacerbated by venous pooling during exercise (exercise-induced), after meals (postprandial hypotension), and after prolonged bed rest (deconditioning). Â

- Â â^{PADHTJ} Drug-induced OH (most common cause of OH):
 - â^{PADNEH}â^{PADNEH} e.g. vasodilators, diuretics, phenothiazine, antidepressants
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- Â â^{PADHTJ} Primary autonomic failure (neurogenic OH):
 - â^{PADNEH}â^{PADNEH} pure autonomic failure, multiple system atrophy, Parkinsonâ^{PADSGC}s disease, dementia with Lewy bodies
- Â â^{PADHTJ} Secondary autonomic failure (neurogenic OH):
 - â^{PADNEH}â^{PADNEH} diabetes, amyloidosis, spinal cord injuries, auto-immune autonomic neuropathy, paraneoplastic autonomic neuropathy, kidney failure

Cardiac syncope

Arrhythmia as primary cause:

- Bradycardia:
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- **Cardiopulmonary and great vessels:** pulmonary embolus, acute aortic dissection, pulmonary hypertension

Remarks

- All forms of syncope, but mostly reflex syncope and OH, are more likely to occur or are more severe when various factors are present: medication causing low BP (due to vasodilatation or hypovolaemia), alcohol use, volume depletion (haemorrhage, low fluid intake, diarrhoea, vomiting), pulmonary diseases causing reduction in brain oxygen supply, environmental factors (thermal stress).
- There are two main pathophysiological mechanisms in reflex syncope. **Vasodepression** refers to conditions in which insufficient sympathetic vasoconstriction results in hypotension.¹ **Cardioinhibition** is used when bradycardia or asystole predominates, reflecting a shift towards parasympathetic predominance. The haemodynamic pattern, i.e. cardioinhibitory, vasodepressive, or both, is independent of the trigger evoking reflex syncope. For example, micturition syncope and orthostatic VVS may equally well present as cardioinhibitory or as vasodepressor syncope
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- The cardiovascular causes of orthostatic intolerance include classical OH, initial OH, delayed OH, POTS, and VVS, which in this context can be called orthostatic VVS.⁶ and ⁷ Syndromes of orthostatic intolerance that may cause syncope are presented in [Web Practical Instruction section 2](#).

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- **orthostatic VVS:** standing, less common sitting
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 - **gastrointestinal stimulation** (swallow, defaecation)

- o ~~â~~~~â~~ cough, sneeze
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 - ~~â~~~~â~~ haemorrhage, diarrhoea, vomiting, etc.
 - Â ~~â~~ Primary autonomic failure (neurogenic OH):
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 - Â ~~â~~ Secondary autonomic failure (neurogenic OH):
 - o ~~â~~~~â~~ diabetes, amyloidosis, spinal cord injuries, auto-immune autonomic neuropathy, paraneoplastic autonomic neuropathy, kidney failure

Cardiac syncope

Arrhythmia as primary cause:

- Bradycardia:
 - o ~~â~~~~â~~ sinus node dysfunction (including bradycardia/tachycardia syndrome)
 - o ~~â~~~~â~~ atrioventricular conduction system disease
- o Â ~~â~~ Tachycardia:
 - ~~â~~~~â~~ supraventricular

- **â€¢** ventricular
- **Â** Structural cardiac: aortic stenosis, acute myocardial infarction/ischaemia, hypertrophic cardiomyopathy, cardiac masses (atrial myxoma, tumours, etc.), pericardial disease/tamponade, congenital anomalies of coronary arteries, prosthetic valve dysfunction
- **Â** Cardiopulmonary and great vessels: pulmonary embolus, acute aortic dissection, pulmonary hypertension

Remarks

- All forms of syncope, but mostly reflex syncope and OH, are more likely to occur or are more severe when various factors are present: medication causing low BP (due to vasodilatation or hypovolaemia), alcohol use, volume depletion (haemorrhage, low fluid intake, diarrhoea, vomiting), pulmonary diseases causing reduction in brain oxygen supply, environmental factors (thermal stress).
- There are two main pathophysiological mechanisms in reflex syncope. **â€¢** Vasodepression¹ refers to conditions in which insufficient sympathetic vasoconstriction results in hypotension. **Â** ² **â€¢** Cardioinhibition³ is used when bradycardia or asystole predominates, reflecting a shift towards parasympathetic predominance. The haemodynamic pattern, i.e. cardioinhibitory, vasodepressive, or both, is independent of the trigger evoking reflex syncope. For example, micturition syncope and orthostatic VVS may equally well present as cardioinhibitory or as vasodepressor syncope
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- The cardiovascular causes of orthostatic intolerance include classical OH, initial OH, delayed OH, POTS, and VVS, which in this context can be called orthostatic VVS.⁶ **Â** ⁷ Syndromes of orthostatic intolerance that may cause syncope are presented in [Web Practical Instruction section 2](#).

BP = blood pressure; OH = orthostatic hypotension; POTS = postural orthostatic tachycardia syndrome; VVS = vasovagal syncope.

The pathophysiological classification centres on a fall in systemic blood pressure (BP) with a decrease in global cerebral blood flow as the defining characteristic of syncope. *Figure 3* shows low BP and global cerebral hypoperfusion as the central final common pathway of syncope. A sudden cessation of cerebral blood flow for as short as 6â€¢8 s can cause complete LOC. A systolic BP of 50â€¢60

mmHg at heart level, i.e. 30–45 mmHg at brain level in the upright position, will cause LOC.⁸ ⁹

Systemic BP is the product of cardiac output and total peripheral resistance; a fall in either can cause syncope. However, in syncope, both mechanisms often act together to a varying degree.

There are three primary causes of a low total peripheral resistance. The first is decreased reflex activity causing vasodilatation through withdrawal of sympathetic vasoconstriction: this is the vasodepressive type of reflex syncope, seen in the outer ring in *Figure 3*. The second is a functional impairment, and the third a structural impairment of the autonomic nervous system, with drug-induced, primary, and secondary autonomic failure in the outer ring. In autonomic failure, there is insufficient sympathetic vasoconstriction in response to the upright position.

There are four primary causes of low cardiac output. The first is a reflex bradycardia, known as cardioinhibitory reflex syncope. The second concerns cardiovascular causes: arrhythmia, structural disease including pulmonary embolism, and pulmonary hypertension. The third is inadequate venous return due to volume depletion or venous pooling. Finally, chronotropic and inotropic incompetence through autonomic failure may impair cardiac output.

Note that these primary mechanisms may interact in different ways: firstly, venous pooling and inadequate venous return is also a factor that can trigger an inappropriate reflex in orthostatic reflex syncope; secondly, a low total peripheral resistance may cause venous pooling of blood below the diaphragm, in turn decreasing venous return and consequently cardiac output.

The three main groups of syncope, i.e. reflex, cardiovascular, and secondary to orthostatic hypotension (OH), are shown outside the rings in *Figure 3*. Both reflex syncope and OH span the two main pathophysiological mechanisms.

 Pathophysiological basis of the classification of syncope.

Figure 3

Pathophysiological basis of the classification of syncope.

ANS = autonomic nervous system; auton. = autonomic; BP = blood pressure; OH = orthostatic hypotension; periph. = peripheral; resist. = resistance.

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3.2.2 Non-syncopeal forms of (real or apparent) transient loss of consciousness

Only those forms of epilepsy in which normal motor control is lost, so patients may fall, are included in *Figure 2*. These are tonic, clonic, tonic-clonic, and atonic generalized seizures, and can be classified as primary or secondary. The forms of epilepsy in which people remain actively upright, i.e. sitting or standing (e.g. complex partial seizures or absence epilepsy) are not regarded as TLOC, but sometimes they are incorrectly diagnosed as syncope.

Psychogenic TLOC consists of two forms: one resembles epileptic seizures (psychogenic non-epileptic seizures [PNES]) and one, without gross movements, resembles syncope (psychogenic pseudosyncope [PPS]).

The rare causes of TLOC only seldomly cause confusion with the main TLOC forms, probably because in most cases they differ enough clinically to be clearly not syncope. Both vertebrobasilar transient ischaemic attacks (TIAs) and subclavian steal syndrome are associated with focal neurological signs. A subarachnoid haemorrhage may present with a short LOC, but the associated abrupt extreme headache suggests the cause. In cyanotic breath-holding spells, expiratory apnoea with hypoxia is the primary mechanism.¹⁰ So-called ‘pallid breath-holding spells’ in children do not constitute a primary respiratory problem, but are cardioinhibitory reflex syncope.¹¹

Table 4 lists the main features that distinguish syncope from disorders that may be mistaken for syncope.

Table 4

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Conditions that may be incorrectly diagnosed as syncope

Condition	Characteristic features that distinguish from syncope
Generalized seizures	See section 8, Table 10.
Complex partial seizures, absence epilepsy	No falls, yet unresponsive and later amnesia
PPS or ‘pseudocoma’	Duration of apparent LOC lasting many minutes to hours; high frequency, up to several times a day
Falls without TLOC	No unresponsiveness or amnesia
Cataplexy	Falls with flaccid paralysis and non-responsive, yet no later amnesia
Intracerebral or subarachnoid haemorrhage	Consciousness may be progressively reduced rather than immediately lost. Accompanying severe headache, other neurological signs
Vertebrobasilar TIA	Always focal neurological signs and symptoms, usually without LOC; if consciousness is lost this usually lasts longer than in TLOC.

Carotid TIA	Consciousness is for all practical purposes not lost in carotid TIAs, but there are pronounced focal neurological signs and symptoms
Subclavian steal syndrome	Associated with focal neurological signs
Metabolic disorders including hypoglycaemia, hypoxia, hyperventilation with hypocapnia	Duration much longer than in TLOC; consciousness may be impaired instead of lost
Intoxication	Duration much longer than in TLOC; consciousness may be impaired instead of lost
Cardiac arrest	LOC yet no spontaneous recovery
Coma	Duration much longer than TLOC

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Subclavian steal syndrome	Associated with focal neurological signs
Metabolic disorders including hypoglycaemia, hypoxia, hyperventilation with hypocapnia	Duration much longer than in TLOC; consciousness may be impaired instead of lost
Intoxication	Duration much longer than in TLOC; consciousness may be impaired instead of lost
Cardiac arrest	LOC yet no spontaneous recovery
Coma	Duration much longer than TLOC

LOC = loss of consciousness; PPS = psychogenic pseudosyncope; TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

Table 4

[Open in new tab](#)

Conditions that may be incorrectly diagnosed as syncope

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Cardiac arrest	LOC yet no spontaneous recovery
Coma	Duration much longer than TLOC

LOC = loss of consciousness; PPS = psychogenic pseudosyncope; TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

4. Diagnostic evaluation and management according to risk stratification

4.1 Initial evaluation

The clinical features characterizing TLOC are usually derived from history taking from patients and eyewitnesses. When a patient first presents with possible TLOC, history taking should first establish whether there was indeed a TLOC. Often, this allows a distinction between the major TLOC groups. The flow diagram for the evaluation of TLOC is shown in *Figure 4*. The initial evaluation should answer key questions:

1. Was the event TLOC?
2. In case of TLOC, is it of syncopal or non-syncopal origin?
3. In case of suspected syncope, is there a clear aetiological diagnosis (see section 4.1.1)?
4. Is there evidence to suggest a high risk of cardiovascular events or death (see section 4.1.2)?

TLOC has four specific characteristics: short duration, abnormal motor control, loss of responsiveness, and amnesia for the period of LOC (for an explanation of the clinical features of TLOC see [Web Table 4](#) in section 4.1 of the [Web Practical Instructions](#)).

TLOC is probably syncope when: (i) there are signs and symptoms specific for reflex syncope, syncope due to OH, or cardiac syncope, and (ii) signs and symptoms specific for other forms of TLOC (head trauma, epileptic seizures, psychogenic TLOC, and/or rare causes) are absent. Practical instructions for history taking are given in sections 3 and 4 of the [Web Practical Instructions](#).

When epileptic seizures or psychogenic attacks are likely, appropriate steps should be taken. By using a detailed clinical history, physicians can differentiate syncope from other forms of TLOC in approximately 60% of cases.¹² For non-syncopal TLOC, refer to sections 7 and 8.



Flow diagram for the initial evaluation and risk stratification of patients with syncope. BP = blood pressure; ECG = electrocardiogram; H&P exam = history and physical examination; TLOC = transient loss of consciousness.

Figure 4

Flow diagram for the initial evaluation and risk stratification of patients with syncope. BP = blood pressure; ECG = electrocardiogram; H&P exam = history and physical examination; TLOC = transient loss of consciousness.

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4.1.1 Diagnosis of syncope

The starting point of the diagnostic evaluation of TLOC of suspected syncopal nature is the initial syncope evaluation, which consists of:

- Careful history taking concerning present and previous attacks, as well as eyewitness accounts, in person or through a telephone interview.
- Physical examination, including supine and standing BP measurements.
- Electrocardiogram (ECG).

Based on these findings, additional examinations may be performed when needed (see section 4.2):

- Immediate ECG monitoring when there is a suspicion of arrhythmic syncope.
- Echocardiogram when there is previous known heart disease, data suggestive of structural heart disease, or syncope secondary to cardiovascular cause.
- Carotid sinus massage (CSM) in patients aged >40 years.
- Head-up tilt testing when there is suspicion of syncope due to OH or reflex syncope.
- Blood tests when clinically indicated, e.g. haematocrit or haemoglobin when haemorrhage is suspected, oxygen saturation and blood gas analysis when hypoxia is suspected, troponin when cardiac ischaemia-related syncope is suspected, or D-dimer when pulmonary embolism is suspected, etc.

Â Even if there is no independent gold/reference standard to diagnose syncope, there is strong consensus that the initial evaluation may lead to certain or highly likely diagnosis when the diagnostic criteria listed in the table of recommendations are met.

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Diagnostic criteria with initial evaluation

 graphic
 graphic

AV = atrioventricular; BBB = bundle branch block; b.p.m. = beats per minute; ECG = electrocardiogram; ICD = implantable cardioverter defibrillator; OH = orthostatic hypotension; SCD = sudden cardiac death; SVT = supraventricular tachycardia; VT = ventricular tachycardia; VVS = vasovagal syncope.

a
Class of recommendation.

b
Level of evidence.

[Open in new tab](#)

Diagnostic criteria with initial evaluation

 graphic
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a
Class of recommendation.

b
Level of evidence.

When a diagnosis is nearly certain or highly likely, no further evaluation is needed, and treatment can be planned. In other cases, the initial evaluation may suggest a diagnosis when the features listed in *Table 5* are present, or otherwise is unable to suggest any diagnosis.

Table 5
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Clinical features that can suggest a diagnosis on initial evaluation

Reflex syncope Â

- Long history of recurrent syncope, in particular occurring before the age of 40 years
- After unpleasant sight, sound, smell, or pain
- Prolonged standing
- During meal
- Being in crowded and/or hot places
- Autonomic activation before syncope: pallor, sweating, and/or nausea/vomiting
- With head rotation or pressure on carotid sinus (as in tumours, shaving, tight collars)
- Absence of heart disease

Syncope due to OH Â

- While or after standing
- Prolonged standing
- Standing after exertion
- Post-prandial hypotension
- Temporal relationship with start or changes of dosage of vasodepressive drugs or diuretics leading to hypotension
- Presence of autonomic neuropathy or parkinsonism

Cardiac syncope Â

- During exertion or when supine
- Sudden onset palpitation immediately followed by syncope
- Family history of unexplained sudden death at young age
- Presence of structural heart disease or coronary artery disease
- ECG findings suggesting arrhythmic syncope:

- Bifascicular block (defined as either left or right BBB combined with left anterior or left posterior fascicular block)
- Other intraventricular conduction abnormalities (QRS duration ≥ 0.12 s)
- Mobitz I second-degree AV block and 1st degree AV block with markedly prolonged PR interval
- Asymptomatic mild inappropriate sinus bradycardia (40–50 b.p.m.) or slow atrial fibrillation (40–50 b.p.m.) in the absence of negatively chronotropic medications
- Non-sustained VT
- Pre-excited QRS complexes
- Long or short QT intervals
- Early repolarization
- ST-segment elevation with type 1 morphology in leads V1–V3 (Brugada pattern)
- Negative T waves in right precordial leads, epsilon waves suggestive of ARVC
- Left ventricular hypertrophy suggesting hypertrophic cardiomyopathy

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ARVC = arrhythmogenic right ventricular cardiomyopathy; AV = atrioventricular; BBB = bundle branch block; b.p.m. = beats per minute; ECG = electrocardiogram; OH = orthostatic hypotension; VT = ventricular tachycardia.

Table 5

[Open in new tab](#)

Clinical features that can suggest a diagnosis on initial evaluation

Reflex syncope ⚡

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- During meal
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- Absence of heart disease

Syncope due to OH ⚡

- While or after standing
- Prolonged standing
- Standing after exertion
- Post-prandial hypotension
- Temporal relationship with start or changes of dosage of vasodepressive drugs or diuretics leading to hypotension
- Presence of autonomic neuropathy or parkinsonism

Cardiac syncope ⚡

- During exertion or when supine
- Sudden onset palpitation immediately followed by syncope
- Family history of unexplained sudden death at young age

- Presence of structural heart disease or coronary artery disease
- ECG findings suggesting arrhythmic syncope:
 - ~~â€¢~~ Bifascicular block (defined as either left or right BBB combined with left anterior or left posterior fascicular block)
 - ~~â€¢~~ Other intraventricular conduction abnormalities (QRS duration ~~â€¢~~0.12 s)
 - ~~â€¢~~ Mobitz I second-degree AV block and 1st degree AV block with markedly prolonged PR interval
 - ~~â€¢~~ Asymptomatic mild inappropriate sinus bradycardia (40~~â€¢~~50 b.p.m.) or slow atrial fibrillation (40~~â€¢~~50 b.p.m.) in the absence of negatively chronotropic medications
 - ~~â€¢~~ Non-sustained VT
 - ~~â€¢~~ Pre-excited QRS complexes
 - ~~â€¢~~ Long or short QT intervals
 - ~~â€¢~~ Early repolarization
 - ~~â€¢~~ ST-segment elevation with type 1 morphology in leads V1~~â€¢~~V3 (Brugada pattern)
 - ~~â€¢~~ Negative T waves in right precordial leads, epsilon waves suggestive of ARVC
 - ~~â€¢~~ Left ventricular hypertrophy suggesting hypertrophic cardiomyopathy

Reflex syncope Â

- Long history of recurrent syncope, in particular occurring before the age of 40 years
- After unpleasant sight, sound, smell, or pain
- Prolonged standing
- During meal
- Being in crowded and/or hot places
- Autonomic activation before syncope: pallor, sweating, and/or nausea/vomiting
- With head rotation or pressure on carotid sinus (as in tumours, shaving, tight collars)
- Absence of heart disease

Syncope due to OH

- While or after standing
- Prolonged standing
- Standing after exertion
- Post-prandial hypotension
- Temporal relationship with start or changes of dosage of vasodepressive drugs or diuretics leading to hypotension
- Presence of autonomic neuropathy or parkinsonism

Cardiac syncope

- During exertion or when supine
- Sudden onset palpitation immediately followed by syncope
- Family history of unexplained sudden death at young age
- Presence of structural heart disease or coronary artery disease
- ECG findings suggesting arrhythmic syncope:
 - Bifascicular block (defined as either left or right BBB combined with left anterior or left posterior fascicular block)
 - Other intraventricular conduction abnormalities (QRS duration ≥ 0.12 s)
 - Mobitz I second-degree AV block and 1° degree AV block with markedly prolonged PR interval
 - Asymptomatic mild inappropriate sinus bradycardia ($40 \leq 50$ b.p.m.) or slow atrial fibrillation ($40 \leq 50$ b.p.m.) in the absence of negatively chronotropic medications
 - Non-sustained VT
 - Pre-excited QRS complexes
 - Long or short QT intervals
 - Early repolarization
 - ST-segment elevation with type 1 morphology in leads V1-V3 (Brugada pattern)

- o  Negative T waves in right precordial leads, epsilon waves suggestive of ARVC
- o  Left ventricular hypertrophy suggesting hypertrophic cardiomyopathy

ARVC = arrhythmogenic right ventricular cardiomyopathy; AV = atrioventricular; BBB = bundle branch block; b.p.m. = beats per minute; ECG = electrocardiogram; OH = orthostatic hypotension; VT = ventricular tachycardia.

4.1.2 Management of syncope in the emergency department based on risk stratification

The management of TLOC of suspected syncopal nature in the ED should answer the following three key questions:

1. Is there a serious underlying cause that can be identified?
2. What is the risk of a serious outcome?
3. Should the patient be admitted to hospital?

Figure 5 shows a flow chart for the management and risk stratification of patients referred to the ED for TLOC suspected to be syncope (modified from Casagrande *et al.* ⁴⁰).

 The management of patients presenting to the emergency department for transient loss of consciousness suspected to be syncope (modified from Casagrande *et al.* ⁴⁰). ED = emergency department; TLOC = transient loss of consciousness.

Figure 5

The management of patients presenting to the emergency department for transient loss of consciousness suspected to be syncope (modified from Casagrande *et al.* ⁴⁰). ED = emergency department; TLOC = transient loss of consciousness.

^aFor example, this includes pulmonary embolism presenting with shortness of breath, pleuritic chest pain, and syncope, but not trauma secondary to syncope.

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Question 1: Is there a serious underlying cause that can be identified in the ED?

Normally the presenting complaint of syncope can be established. The primary aim for an ED clinician is then to establish an underlying diagnosis, especially those associated with the potential for rapid clinical deterioration.⁴¹ ⁴² It is the acute underlying disease that most frequently determines short-term adverse events rather than the syncope itself.⁴³ Subsequent management will focus on treating this underlying cause (Figure 5). Many (40^a–45%) non-cardiovascular and some cardiovascular life-threatening underlying conditions are obvious in the ED.⁴⁴ Table 6 lists high-risk features that suggest the presence of a serious underlying cause and low-risk features that suggest a benign underlying cause.

Table 6

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High-risk features (that suggest a serious condition) and low-risk features (that suggest a benign condition) in patients with syncope at initial evaluation in the emergency department

 graphic
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 graphic
 graphic

Table 6

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High-risk features (that suggest a serious condition) and low-risk features (that suggest a benign condition) in patients with syncope at initial evaluation in the emergency department

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Question 2: What is the risk of a serious outcome?

High-risk features are shown in *Table 6*, and how to use this risk profile to guide subsequent management and disposition is shown in *Figure 6*.

Risk stratification is important, for two reasons:

1. To recognize patients with a likely *low-risk* condition able to be discharged with adequate patient education.
2. To recognize patients with a likely *high-risk* cardiovascular condition requiring urgent investigation. This may require admission.

High-risk patients are more likely to have cardiac syncope. Structural heart disease^{25a, 27, 31} , ³⁵ , ³⁶ , ⁴⁵ and primary electrical disease⁴⁶ are major risk factors for sudden cardiac death (SCD) and overall mortality in patients with syncope. Low-risk patients are more likely to have

reflex syncope and have an excellent prognosis.⁴⁷ OH is associated with a two-fold higher risk of death owing to the severity of comorbidities compared with the general population.⁴⁸

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Management of syncope in the emergency department

 graphic

 graphic

ED = emergency department; OH = orthostatic hypotension.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Management of syncope in the emergency department

 graphic

 graphic

ED = emergency department; OH = orthostatic hypotension.

a

Class of recommendation.

b

Level of evidence.

 Emergency department risk stratification flow chart. Low- and high-risk features are listed in Table 6. ED = emergency department; SU = syncope unit.

Figure 6

Emergency department risk stratification flow chart. Low- and high-risk features are listed in [Table 6](#).

ED = emergency department; SU = syncope unit.

Patients with low-risk features. These patients do not need further diagnostic tests in the ED as they are likely to have reflex, situational, or orthostatic syncope. They may benefit from reassurance, or

counselling (see [Web Practical Instructions section 9.1: ESC information sheet for patients affected by reflex syncope](#)).

Patients with high-risk features. These patients should be classified as HIGH RISK; they require an intensive diagnostic approach and may need urgent treatment and admission. These patients should be monitored (although it is unclear for how long this should be, most studies suggesting up to 6 hours in the ED and up to 24 hours in hospital) in a setting where resuscitation can be performed in case of deterioration.⁴⁰ ⁶²

Patients that have neither high- nor low-risk features. These patients will require expert syncope opinion, which can probably be safely managed in an outpatient setting.⁶³ There is no direct evidence that admitting patients to hospital changes their outcome, whilst there is evidence that management in an ED observation unit and/or fast-track to a syncope outpatient unit is beneficial.⁶⁴ ⁶⁵

^aRecent studies have suggested that outcomes in patients presenting with presyncope are similar to those presenting with syncope.^{66a} ⁶⁸

^bThese patients may still require admission to hospital for associated illness, injury or welfare reasons. Low-risk patients can be referred to the outpatient syncope clinic for therapy purposes, if needed.

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Question 3: Should the patient be admitted to hospital?

Approximately 50% of patients who present to the ED with syncope are admitted (although the rate varies between 12%⁶⁹–86%) (see [Supplementary Data Table 4](#)). The use of clinical decision rules and standardized protocols has not changed this rate significantly. The composite estimate of outcomes is that in the next 7%⁷⁰–30 days, only 0.8% die and 6.9% have a non-fatal severe outcome whilst in the ED, while another 3.6% have a post-ED serious outcome (see [Supplementary Data Table 4](#)).

Unnecessary admission in low-risk patients can be harmful.⁸⁷ Whereas it is crucial to identify these high-risk patients to ensure early, rapid, and intensive investigation, not all patients at high risk need hospitalization.⁸⁰

The diagnostic tests, procedures, and interventions that may require admission in patients with high-risk features are listed in [Table Z](#). Furthermore, this Task Force believes that the implementation of novel care pathways and organizational approaches, such as ED observation units and syncope in- and outpatient units ([Figure 6](#)), offer safe and effective alternatives to admission in the cases listed in [Table Z](#). Based on a consensus document,⁴⁰ a single-centre experience consisting of a short stay in the ED under observation of a⁷¹–48 h coupled with fast-tracking to a syncope unit reduced the admission rate to 29%.⁷⁷ Among patients not admitted, 20% were discharged after a short observation in the ED, 20% were fast-tracked to the syncope unit, and 31% were discharged directly from the ED.⁷⁷

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High-risk syncope patients: criteria favouring a stay in an emergency department observation unit and/or fast-tracking to a syncope unit vs. requiring admission to hospital

Favour initial management in ED observation unit and/or fast-track to syncope unit	Favour admission to hospital
High-risk features AND: ⚠ • Stable, known structural heart disease • Severe chronic disease • Syncope during exertion • Syncope while supine or sitting • Syncope without prodrome • Palpitations at the time of syncope • Inadequate sinus bradycardia or sinoatrial block • Suspected device malfunction or inappropriate intervention • Pre-excited QRS complex • SVT or paroxysmal atrial fibrillation • ECG suggesting an inheritable arrhythmogenic disorders • ECG suggesting ARVC	High-risk features AND: ⚠ • Any potentially severe coexisting disease that requires admission • Injury caused by syncope • Need of further urgent evaluation and treatment if it cannot be achieved in another way (i.e. observation unit), e.g. ECG monitoring, echocardiography, stress test, electrophysiological study, angiography, device malfunction, etc. • Need for treatment of syncope
Favour initial management in ED observation unit and/or fast-track to syncope unit	Favour admission to hospital
High-risk features AND: ⚠	High-risk features AND: ⚠

• Stable, known structural heart disease • Severe chronic disease • Syncope during exertion • Syncope while supine or sitting • Syncope without prodrome • Palpitations at the time of syncope • Inadequate sinus bradycardia or sinoatrial block • Suspected device malfunction or inappropriate intervention • Pre-excited QRS complex • SVT or paroxysmal atrial fibrillation • ECG suggesting an inheritable arrhythmogenic disorders • ECG suggesting ARVC	• Any potentially severe coexisting disease that requires admission • Injury caused by syncope • Need of further urgent evaluation and treatment if it cannot be achieved in another way (i.e. observation unit), e.g. ECG monitoring, echocardiography, stress test, electrophysiological study, angiography, device malfunction, etc. • Need for treatment of syncope
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ARVC = arrhythmogenic right ventricular cardiomyopathy; ECG = electrocardiogram; ED = emergency department; SVT = supraventricular tachycardia.

Table 7

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High-risk syncope patients: criteria favouring a stay in an emergency department observation unit and/or fast-tracking to a syncope unit vs. requiring admission to hospital

Favour initial management in ED observation unit and/or fast-track to syncope unit	Favour admission to hospital
High-risk features AND: Â • Stable, known structural heart disease	High-risk features AND: Â • Any potentially severe coexisting disease that requires admission • Injury caused by syncope

- | | |
|--|---|
| <ul style="list-style-type: none"> • Severe chronic disease • Syncope during exertion • Syncope while supine or sitting • Syncope without prodrome • Palpitations at the time of syncope • Inadequate sinus bradycardia or sinoatrial block • Suspected device malfunction or inappropriate intervention • Pre-excited QRS complex • SVT or paroxysmal atrial fibrillation • ECG suggesting an inheritable arrhythmogenic disorders • ECG suggesting ARVC | <ul style="list-style-type: none"> • Need of further urgent evaluation and treatment if it cannot be achieved in another way (i.e. observation unit), e.g. ECG monitoring, echocardiography, stress test, electrophysiological study, angiography, device malfunction, etc. • Need for treatment of syncope |
|--|---|

Favour initial management in ED observation unit and/or fast-track to syncope unit	Favour admission to hospital
High-risk features AND: Â • Stable, known structural heart disease • Severe chronic disease • Syncope during exertion • Syncope while supine or sitting • Syncope without prodrome • Palpitations at the time of syncope	High-risk features AND: Â • Any potentially severe coexisting disease that requires admission • Injury caused by syncope • Need of further urgent evaluation and treatment if it cannot be achieved in another way (i.e. observation unit), e.g. ECG monitoring, echocardiography, stress test, electrophysiological study, angiography, device malfunction, etc. • Need for treatment of syncope

- Inadequate sinus bradycardia or sinoatrial block
- Suspected device malfunction or inappropriate intervention
- Pre-excited QRS complex
- SVT or paroxysmal atrial fibrillation
- ECG suggesting an inheritable arrhythmogenic disorders
- ECG suggesting ARVC

ARVC = arrhythmogenic right ventricular cardiomyopathy; ECG = electrocardiogram; ED = emergency department; SVT = supraventricular tachycardia.

Risk stratification scores:

There are several ED syncope clinical decision rules that aim to stratify patients with syncope based on medical history, examination, and ECG findings (see [Supplementary Data Table 3](#)).²⁶ ³⁴ ³⁶ ⁴⁴ ⁸⁸ None of these rules are used widely in EDs due to poor sensitivity and specificity reported from external validation, or due to a lack of external validation.⁷⁰ ⁷⁸ ⁸⁵ Syncope clinical decision rules perform no better than clinician judgment at predicting short-term serious outcomes.⁸⁶ Clinical decision rules can predict poor outcomes, but most syncope deaths and many poor outcomes are associated with underlying illness rather than syncope *per se*,⁵⁸ particularly in the long term.⁵⁶

Even if the quality of evidence is moderate, there is strong consensus from several studies that currently available risk stratification scores have not shown better sensitivity, specificity, or prognostic yield compared with clinical judgment in predicting short-term serious outcomes after syncope. Therefore, they should not be used alone to perform risk stratification in the ED.

4.2 Diagnostic tests

4.2.1 Carotid sinus massage

A ventricular pause lasting >3 s and/or a fall in systolic BP of >50 mmHg is known as carotid sinus hypersensitivity. Carotid sinus hypersensitivity is a common finding in older men without syncope; abnormal responses are frequently observed (approximately 40%) in patients without syncope, especially if they are older and affected by cardiovascular disease.⁸⁹ Carotid sinus hypersensitivity is exceptional in patients <40 years of age.⁹⁰ The specificity of the test increases if spontaneous syncope is reproduced

during CSM. Syncope was induced in only 5% of asymptomatic persons aged >65 years.⁸⁹ For the above reasons, the diagnosis of carotid sinus syndrome (CSS) requires the reproduction of spontaneous symptoms and, in addition, that patients have syncope of unknown origin compatible with a reflex mechanism. In such circumstances, CSM usually shows a period of asystole >6 s.⁹¹ The prevalence of CSS, as defined here, was 8.8% when CSM was performed after the initial evaluation in 1855 consecutive patients >40 years of age with syncope compatible with a reflex mechanism.⁹² In a multicentre study⁹⁴ aimed at validation of the 2009 ESC Guidelines, CSM was indicated after initial evaluation in 73% of 700 patients and was diagnostic in 12%. The precise methodology and results of CSM are shown in section 5 of the [Web Practical Instructions](#).

The main complications of CSM are neurological. When pooling the data from four studies⁹⁰ in which 8720 patients were analysed, TIAs or strokes were observed in 21 (0.24%).

The relationship between abnormal response to CSM and spontaneous syncope is a crucial point that has been studied using two methods. The first was a preâpost comparison of the recurrence rate of syncope after pacing. Non-randomized studies demonstrated fewer recurrences at follow-up in paced patients than in those without pacing. These results were confirmed in two randomized trials.⁹⁸ The second method was to analyse the occurrence of asystolic episodes registered in patients with a cardioinhibitory response to CSM using an implanted device. Recordings of long pauses were very common in the two trials that employed this method.¹⁰⁰ These results suggest that a positive response to CSM, reproducing symptoms, in patients with syncope is highly predictive of the occurrence of spontaneous asystolic episodes.

There is strong consensus that the diagnosis of CSS requires both the reproduction of spontaneous symptoms during CSM and clinical features of spontaneous syncope compatible with a reflex mechanism. The quality of evidence is moderate and is given by studies of ECG correlation between CSM and spontaneous events, and indirectly by studies of efficacy of cardiac pacing. Further research is likely to have an important impact on our confidence in the estimation of effect and may change the estimate.

[Open in new tab](#)

Cardiac sinus massage

 graphic

 graphic

BP = blood pressure; CSM = carotid sinus massage; CSS = carotid sinus syndrome; TIA = transient ischaemic attack.

a

Class of recommendation.

b

Level of evidence.

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Cardiac sinus massage



BP = blood pressure; CSM = carotid sinus massage; CSS = carotid sinus syndrome; TIA = transient ischaemic attack.

a

Class of recommendation.

b

Level of evidence.

4.2.2 Orthostatic challenge

Changing from the supine to the upright position produces a displacement of blood from the thorax to the lower limbs and abdominal cavity that leads to a decrease in venous return and cardiac output. In the absence of compensatory mechanisms, a fall in BP may lead to syncope.²⁰ ¹⁰³ ¹⁰⁴ The diagnostic criteria for OH have been defined by consensus.⁶

Currently, there are three methods for assessing the response to change in posture from supine to erect²⁰ ¹⁰³ ¹⁰⁴: active standing (see section 4.2.2.1), head-up tilt (see section 4.2.2.2), and 24-h ambulatory BP monitoring (ABPM) (see section 4.2.3.4).

4.2.2.1 Active standing

Indications:

This test is used to diagnose different types of orthostatic intolerance (see [Web Practical Instructions Web Table 1](#)). A sphygmomanometer is adequate for routine clinical testing for classical OH and delayed OH because of its ubiquity and simplicity. Automatic arm-cuff devices, which are programmed to repeat and confirm measurements when discrepant values are recorded, are at a disadvantage due to the rapidly falling BP during OH. With a sphygmomanometer, more than four measurements per minute cannot be obtained without venous obstruction in the arm. When more frequent readings are required, as for initial OH, continuous beat-to-beat non-invasive BP measurement is needed.²⁰ ¹⁰³ ¹⁰⁴

Diagnostic criteria:

Abnormal BP fall is defined as a progressive and sustained fall in systolic BP from baseline value ≥ 20 mmHg or diastolic BP ≥ 10 mmHg, or a decrease in systolic BP to <90 mmHg. This definition of OH differs from the 2011 consensus⁶ in adding the 90 mmHg threshold. This Task Force believes that an absolute threshold of 90 mmHg of systolic BP is useful, especially in patients with a supine BP <110 mmHg. An isolated diastolic BP drop is very rare and its clinical relevance for OH diagnosis is limited. Orthostatic heart rate (HR) increase is blunted or absent [usually not >10 beats per minute (b.p.m.)] in patients with neurogenic OH, but increases or even exaggerates with anaemia or hypovolaemia. The probability that syncope and orthostatic complaints are due to OH can be assessed using the information given in *Table 8*.

Table 8

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Association of orthostatic intolerance and orthostatic hypotension



BP = blood pressure; OH = orthostatic hypotension.

Table 8

[Open in new tab](#)

Association of orthostatic intolerance and orthostatic hypotension



BP = blood pressure; OH = orthostatic hypotension.

[Open in new tab](#)

Active standing



BP = blood pressure; b.p.m. = beats per min; OH = orthostatic hypotension; HR = heart rate; POTS = postural orthostatic tachycardia syndrome.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Active standing



BP = blood pressure; b.p.m. = beats per min; OH = orthostatic hypotension; HR = heart rate; POTS = postural orthostatic tachycardia syndrome.

a

Class of recommendation.

b

Level of evidence.

4.2.2.2 Tilt testing

Since its introduction in 1986,¹⁰⁵ many protocols have been reported with variations in the initial stabilization phase, duration, tilt angle, type of support, and pharmacological provocation. The most commonly used are the trinitroglycerin (TNG) test using 300–400 μg of sublingual TNG after a 20-min unmedicated phase,¹⁰⁶ and the low-dose intravenous isoproterenol test, which uses incremental doses to increase average HR by about 20–25% over baseline (usually 3 $\mu\text{g}/\text{min}$).¹⁰⁸ In a recent systematic literature review,¹¹⁰ the overall positivity rate in patients with syncope was 66% for the TNG protocol and 61% for the isoproterenol protocol; the respective positivity rate in subjects without syncope (controls) ranged from 11–14%; and the test differentiated patients with syncope from controls with an odds ratio of 12. The methodology and classification of responses are described in section 6 of the [Web Practical Instructions](#). Adding video recording to a tilt table permits objective and repeated review of clinical signs in relation to BP and HR, and helps to assess the relative contribution of bradycardia and hypotension to syncope (see section 5.2.6.3 and the explanatory video in [Web Practical Instructions](#) section 6.3.15), and to distinguish between VVS and PPS (see section 4.2.5).

The clinical situation corresponding to tilt-induced syncope is that which is triggered by prolonged standing. The test should be considered: (i) to confirm a diagnosis of reflex syncope in patients in whom this diagnosis was suspected but not confirmed by initial evaluation^{105–109} and (ii) for the assessment of autonomic failure, especially for the reproduction of delayed OH (which

could not be detected by active standing because of its delayed onset)²³ & ²⁴ & ¹¹² & ¹¹³ and postural orthostatic tachycardia syndrome (POTS).¹¹⁴ Tilt testing may be helpful in separating syncope from PPS.¹¹⁵^{PADSTS}¹¹⁷

Tilt testing has limited value in assessing treatment efficacy.¹¹⁸ However, tilt testing is widely accepted as a useful tool to demonstrate susceptibility of the patient to reflex syncope, especially a hypotensive (vasodepressive) tendency, and thereby to initiate treatment (e.g. physical manoeuvres, see section 5).¹¹⁹^{PADSTS}¹²¹

The endpoint of tilt testing is the reproduction of symptoms along with the characteristic circulatory pattern of the indication mentioned above, namely the induction of reflex hypotension/bradycardia, OH, POTS, or PPS. The typical tilt test result patterns are shown in the [Web Practical Instructions](#) section 6.

Interpretation of tilt testing results in patients with reflex syncope:

Some studies¹²² & ¹²³ compared the response to tilt testing with spontaneous syncope recorded by an implantable loop recorder (ILR). While a positive cardioinhibitory response to tilt testing predicts, with a high probability, an asystolic spontaneous syncope, the presence of a positive vasodepressor, mixed response, or even a negative response, does not exclude the presence of asystole during spontaneous syncope.¹²² & ¹²³

Tilt testing has an acceptable sensitivity¹²⁴ and specificity¹⁰⁶ & ¹²⁴ & ¹²⁵ when these are calculated in patients with true VVS or without a history of syncope. However, there is an inability to apply the test to populations with syncope of uncertain cause where it is hoped that tilt testing might prove decisive. In these clinical settings, tilt testing fails to deliver (*Figure 2*). Indeed, tilt testing was positive in 51^{PADSTS}56% of patients with atypical clinical features suggesting a reflex mechanism,¹⁰⁶ & ¹²⁴^{PADSTS}¹²⁸ in 30^{PADSTS}36% with unexplained syncope after full investigation,¹²⁴ & ¹²⁹ and in 45^{PADSTS}47% with true cardiac arrhythmic syncope.¹³⁰ & ¹³¹ In other words, tilt testing offers little diagnostic value in patients for whom it is most needed. In these patients, a positive tilt test reveals a susceptibility to orthostatic stress.¹³² This *hypotensive susceptibility* plays a role in causing syncope irrespective of the aetiology and mechanism of syncope. For example, in arrhythmic syncope caused by paroxysmal atrial tachyarrhythmias, the mechanism is a combination of onset of the arrhythmia itself and hypotensive susceptibility, corroborated by positive tilt testing.¹³⁰ & ¹³¹ Similarly, multifactorial mechanisms are likely in other types of cardiac syncope, e.g. aortic stenosis,¹³³ hypertrophic cardiomyopathy (HCM),¹³⁴ and sick sinus syndrome.¹³⁵ & ¹³⁶ The presence or absence of susceptibility explains the occurrence of syncope in some and not in others affected by the same severity of arrhythmia or structural defect. Tilt testing should now be considered a means of exposing a hypotensive tendency rather than being diagnostic of VVS. This concept has practical implications for therapy (see sections 5.1 and 5.2).

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Tilt testing



EEG = electroencephalogram; OH = orthostatic hypotension; POTS = postural orthostatic tachycardia syndrome; PPS = psychogenic pseudosyncope; VVS = vasovagal syncope.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Tilt testing



EEG = electroencephalogram; OH = orthostatic hypotension; POTS = postural orthostatic tachycardia syndrome; PPS = psychogenic pseudosyncope; VVS = vasovagal syncope.

a

Class of recommendation.

b

Level of evidence.

 Rates of tilt testing positivity in different clinical conditions. These studies used the Westminster protocol for passive tilt,¹²⁵ the Italian protocol for trinitroglycerin tilt,¹⁰⁶ and the clomipramine protocol,¹²⁴ for a total of 1453 syncope patients and 407 controls without syncope. Studies using other tilt protocols, e.g. isoproterenol challenge, were not included. Clom = clomipramine; TNG = trinitroglycerin; VVS = vasovagal syncope.

Figure 7

Rates of tilt testing positivity in different clinical conditions. These studies used the Westminster protocol for passive tilt,¹²⁵ the Italian protocol for trinitroglycerin tilt,¹⁰⁶ and the clomipramine protocol,¹²⁴ for a total of 1453 syncope patients and 407 controls without syncope. Studies using other

tilt protocols, e.g. isoproterenol challenge, were not included. Clom = clomipramine; TNG = trinitroglycerin; VVS = vasovagal syncope.

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4.2.3 Basic autonomic function tests

Autonomic function assessment helps to identify autonomic failure as the underlying cause of syncope.

4.2.3.1 Valsalva manoeuvre

The methodology of the Valsalva manoeuvre is described in section 7.1.1 of the [Web Practical Instructions](#) and in [Web Video 2](#). There is strong evidence that the absence of a BP overshoot and an absence of a HR increase during the Valsalva is pathognomonic for neurogenic OH, occurring in primary and secondary autonomic failure, and the degree of hypotension and/or lack of compensation during forced expiration usually correlate with the degree of autonomic dysfunction and related symptoms.^{138a}¹⁴³ In contrast, a pronounced BP fall beyond what is normally expected during forced expiration, but a normal chronotropic response during the manoeuvre, may occur in patients with suspected situational syncope, i.e. syncope occurring during some forms of situational syncope, e.g. coughing, brass instrument playing, singing, and weightlifting.¹⁴⁴

4.2.3.2 Deep breathing

The methodology of the deep-breathing test is described in section 7.1.2 of the [Web Practical Instructions](#). Under physiological conditions, HR rises during inspiration and falls during expiration. HR variability during deep breathing (also called the expiratory/inspiratory index or E/I index) is ≈ 15 b.p.m. in healthy individuals aged >50 years.¹⁴⁵ There is strong consensus that blunted or abolished variation is suggestive of parasympathetic dysfunction.¹⁴² ¹⁴³ ¹⁴⁶ ¹⁴⁷

4.2.3.3 Other autonomic function tests

Further tests to evaluate cardiovascular sympathetic function include calculation of the 30:15 ratio, the cold pressure test, the sustained hand grip test, and mental arithmetic. There is weak evidence that these tests may be useful.¹³ ¹⁴² ¹⁴³ ¹⁴⁷

4.2.3.4 Twenty-four-hour ambulatory and home blood pressure monitoring

Twenty-four-hour ABPM and home BP monitoring (HBPM) are increasingly used to diagnose and monitor the treatment of hypertension.¹⁴⁸ There is strong evidence that OH is frequently associated with a nocturnal ¹⁴⁹ non-dipping¹⁵⁰ or even ¹⁵¹ reverse-dipping¹⁵² BP pattern in patients with autonomic failure, with relevant therapeutic and prognostic implications¹⁴⁰ ^{148a}¹⁵¹ (see [Web Practical Instructions](#) section 7.1.3). In these patients, ABPM allows the assessment of nocturnal hypertension, postprandial hypotension, and exercise- and drug-induced hypotension, as well as monitoring for side effects of antihypertensive regimens and pointing to additional disorders such as sleep apnoea.¹⁵² There is weak evidence that ABPM may also detect the degree of OH in daily life better than single office BP measurements.¹⁵³

HBPM may be used to investigate the cause of orthostatic intolerance, i.e. to clarify whether symptoms are due to OH or to other causes, such as vertigo or motor imbalance in Parkinson's disease or multiple system atrophy. The evidence is weak. Finally, HBPM can be used to clarify that BP is not low during episodes of PPS.¹⁵⁴

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Basic autonomic function tests



ABPM = ambulatory blood pressure monitoring; BP = blood pressure; HBPM = home blood pressure monitoring; HR = heart rate; OH = orthostatic hypotension.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Basic autonomic function tests



ABPM = ambulatory blood pressure monitoring; BP = blood pressure; HBPM = home blood pressure monitoring; HR = heart rate; OH = orthostatic hypotension.

a

Class of recommendation.

b

Level of evidence.

4.2.4 Electrocardiographic monitoring (non-invasive and invasive)

The role of ECG monitoring cannot be defined in isolation. As a rule, ECG monitoring is indicated only when there is a high pre-test probability of identifying an arrhythmia associated with syncope, such as those listed in *Table 5*.

4.2.4.1 In-hospital monitoring

In-hospital monitoring (in bed or by telemetry) is warranted in patients with high-risk clinical features (defined in [Table 6](#)) suggesting arrhythmic syncope, especially if the monitoring is applied immediately after syncope. Although the diagnostic yield of ECG monitoring varies from 1.9%¹⁵⁸ to 17.6%,¹⁶⁰ it is justified by the need to avoid immediate risk to the patient.

4.2.4.2 Holter monitoring

Since, in most patients, symptoms do not recur during monitoring, the true yield of Holter monitoring in syncope may be as low as 1%¹⁶¹ in an unselected population. In 15% of patients, symptoms were not associated with arrhythmia.¹⁶¹ Thus, in these patients, a rhythm disturbance could potentially be excluded as a cause of syncope. Holter monitoring in syncope is inexpensive in terms of set-up costs, but expensive in terms of cost per diagnosis.¹⁶² Holter monitoring in syncope may be of more value if symptoms are frequent. Daily single or multiple episodes of LOC might increase the potential for symptomatic ECG correlation.

4.2.4.3 Prospective external event recorders

Event recorders are external devices that are applied by the patient when symptoms occur. Whereas these recorders can be useful in the investigation of palpitations,¹⁶³ they have a marginal role in the evaluation of syncope.

4.2.4.4 Smartphone applications

Because up to now smartphone applications have recorded real-time ECG, their current role in syncope is limited for the same reason as for prospective event recorders.¹⁶⁴ However, home video records are very useful in all forms of TLOC (see section 4.2.5.2).

4.2.4.5 External loop recorders

In general, external loop recorders have a higher diagnostic yield than Holter monitoring.¹⁶² External loop recorders can be useful in patients with relatively frequent syncope episodes.¹⁶⁶ In a recent multicentre international registry, the diagnostic yield in syncope was 24.5%, with the most common finding being bradyarrhythmias; the stronger predictor for diagnostic findings was early monitoring after the index event.¹⁶⁶

4.2.4.6 Remote (at home) telemetry

Most recently, external and implantable device systems have been developed that provide continuous ECG recording or 24-h loop memory with wireless transmission (real time) to a service centre. Some recent studies have shown that implementing remote monitoring increases the diagnostic yield and achieves diagnosis earlier than without remote monitoring.¹⁶⁹

4.2.4.7 Implantable loop recorders

In a meta-analysis of five randomized controlled trials (RCTs),¹⁷² 660 patients with unexplained syncope were randomized to a conventional strategy consisting of an external loop recorder, tilt testing, and an electrophysiological study (EPS), or to prolonged monitoring with an ILR.

The results showed that initial implantation of an ILR in the workup provided a 3.7 [95% confidence interval (CI) 2.7–5.0] increased relative probability of a diagnosis compared with the conventional strategy (see [Supplementary Data Table 5](#)). ILR was more cost-effective than a conventional strategy.¹⁷² ¹⁷³ ¹⁷⁷ ¹⁷⁸

In pooled data from nine studies¹⁷⁹ performed in 506 patients with unexplained syncope at the end of complete negative work-up, a correlation between syncope and ECG was found in 176 patients (35%); of these, 56% had asystole (or bradycardia in a few cases) at the time of the recorded event, 11% had tachycardia, and 33% had no arrhythmia. Presyncope was much less likely to be associated with an arrhythmia than syncope. Similar findings were subsequently observed with ILR use expanded in an early phase of evaluation in patients with recurrent syncope of uncertain origin, in the absence of high-risk criteria and structural heart disease,¹⁷⁶ ¹⁸⁰¹⁸³ and in suspected reflex syncope.¹⁸⁴¹⁸⁶ In particular, an asystolic pause was present during syncope in about 50% of these patients.

There are several areas of interest other than unexplained syncope in which ILRs have been investigated:

- Patients with bundle branch block (BBB) in whom paroxysmal atrioventricular (AV) block is likely despite negative complete EPS: an arrhythmia was observed in 41% of these patients (being paroxysmal AV block in 70%) under ILR observation, based on pooled data from three studies¹⁷⁴ ¹⁸⁷ ¹⁸⁸ (see [Supplementary Data Table 6](#)).
- Patients in whom epilepsy was suspected but the treatment has proven ineffective: in pooled data, an attack could have been documented by ILR in 62% of patients, with an arrhythmic cause being responsible in 26%¹³⁷ ¹⁸⁹¹⁹¹ (see [Supplementary Data Table 7](#)).
- Patients with unexplained falls: in pooled data, an attack could have been documented by ILR in 70% of patients, with an arrhythmic cause being responsible in 14%¹⁹¹¹⁹⁴ (see [Supplementary Data Table 8](#)).
- Patients with HCM, arrhythmogenic right ventricular cardiomyopathy (ARVC), or primary electrical diseases (see section 5.4).

4.2.4.8 Diagnostic criteria

The gold standard for the diagnosis of arrhythmic syncope is when there is a correlation between the symptoms and an ECG recording.¹⁹⁵ ¹⁹⁶ The presence of asymptomatic significant arrhythmias¹⁹⁷ defined as prolonged asystole (≥ 3 s), rapid supraventricular tachycardias (SVTs) (i.e. >160 b.p.m. for >32 beats), or ventricular tachycardias (VTs)¹⁹⁸ has been considered by several authors to be a diagnostic finding.¹⁸⁵ ¹⁸⁸ ¹⁹⁷¹⁹⁹ On the other hand, although the absence of documentation of an arrhythmia during a syncopal episode cannot be considered to be a

specific diagnosis, it allows the exclusion of an arrhythmia as the mechanism of the syncope. Most evidence in support of the above diagnostic criteria is indirectly based on the benefit of specific therapies guided by ECG monitoring in preventing syncopal recurrences.¹⁷² , ¹⁸⁴ ¹⁸⁶ , ¹⁸⁸ , ²⁰⁰

Even if the quality of evidence is moderate, there is strong consensus based on evidence from several controlled trials that a correlation between symptoms and a documented arrhythmia, or the presence of some asymptomatic significant arrhythmias (defined above), is diagnostic of the cause of syncope and specific treatment must be prescribed.

The principal limitation of any ECG monitoring device is the inability to record BP together with ECG. In reflex syncope, the documentation of bradycardia/asystole during a syncopal episode does not rule out the possibility that a hidden hypotensive reflex is the principal cause for syncope, and that bradycardia/asystole is a secondary late event. This issue has important implications for therapy (see section 5). A classification of ECG recordings with their probable related pathophysiology is available in [Web Table 6](#) and [Web Practical Instructions](#) section 8.

[Open in new tab](#)

Electrocardiographic monitoring





AV = atrioventricular; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

a
Class of recommendation.

b
Level of evidence.

[Open in new tab](#)

Electrocardiographic monitoring





AV = atrioventricular; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

a

Class of recommendation.

b

Level of evidence.

4.2.5 Video recording in suspected syncope

4.2.5.1 In-hospital video recording

For PNES, a video-electroencephalogram (EEG) forms the highest level of diagnostic probability.²⁰⁴ For syncope and PPS, video can play a similar, probably underused, role (see section 7). Adding video recording to a tilt table test adds the ability to review clinical signs in relation to BP and HR objectively and repeatedly, thus helping to distinguish VVS from PPS. This approach has revealed new pathophysiological insights in syncope.⁹ Attaching the camera to the tilt table allows detailed study of the face and head, which is useful when assessing the start and end of LOC.⁹ · · · ²⁰⁵ Video recording of tilt-induced PPS¹¹⁶ ensures that apparent TLOC occurs while BP and HR are not low; adding an EEG increases the diagnostic probability of PPS even further. The method has been proven to show the combined presence of VVS and PPS.¹¹⁷

4.2.5.2 Home video recording

Home video records (by means of smartphone technology) are very useful in all forms of TLOC to allow signs of an attack to be studied. Patients and their relatives should be urged to record attacks, if possible, in cases of diagnostic uncertainty. In epilepsy, advances are made towards prolonged video and EEG recording in patients' homes.²⁰⁶ · · · ²⁰⁷ For syncope or PPS, experience suggests that the chances of obtaining a video record are higher for PPS than for syncope, which is probably the effect of a high frequency and long duration of attacks in PPS. It is rare for the beginning of events to be recorded.²⁰⁶ Home video records allow complex events such as syncope-induced epileptic seizures to be diagnosed.²⁰⁸

[Open in new tab](#)

Video recording in suspected syncope



a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Video recording in suspected syncope



a

Class of recommendation.

b

Level of evidence.

4.2.6 Electrophysiological study

Indications:

In an overview of eight studies, including 625 patients with syncope undergoing EPS,²⁰⁹ positive results occurred predominantly in patients with structural heart disease. In recent years, the development of powerful non-invasive methods, i.e. prolonged ECG monitoring, showing a higher diagnostic value, has decreased the importance of EPS as a diagnostic test. In clinical practice, registry data show that approximately 3% of patients with unexplained syncope evaluated by cardiologists undergo EPS and even fewer if they are evaluated by other specialists.⁷¹ Nevertheless, EPS remains useful for diagnosis in the following specific clinical situations: asymptomatic sinus bradycardia (suspected sinus arrest causing syncope), bifascicular BBB (impending high-degree AV block), and suspected tachycardia.

Diagnostic criteria:

Â

4.2.6.1 Asymptomatic sinus bradycardia: suspected sinus arrest causing syncope

The pre-test probability of bradycardia-related syncope is relatively high when there is asymptomatic sinus bradycardia (<50 b.p.m.) or sinoatrial block, usually documented by 12-lead ECG or ECG monitoring. The prognostic value of a prolonged sinus node recovery time (SNRT) is not well defined. An abnormal response is defined as ≥ 1.6 or 2 s for SNRT, or ≥ 525 ms for corrected SNRT.²¹⁰ One observational study showed a relationship between the presence of prolonged SNRT at EPS and the effect of pacing on symptoms.²¹¹ Another small prospective study showed that a corrected SNRT ≥ 800 ms had an eight-fold higher risk of syncope than a SNRT below this value.²¹²

4.2.6.2 Syncope in bifascicular bundle branch block (impending high-degree atrioventricular block)

Patients with bifascicular block and syncope are at higher risk of developing high-degree AV block.²¹³ A prolonged HV interval ≥ 70 ms, or induction of second- or third-degree AV block by pacing or by pharmacological stress (ajmaline, procainamide, or disopyramide), identifies a group at higher risk of developing AV block. By combining the above-mentioned parts of the electrophysiological protocol, a positive EPS yielded a positive predictive value as high as $\geq 80\%$ for the identification of patients who will develop AV block in old studies.^{214,215,216} This finding has been indirectly confirmed by recent studies that showed a significant reduction in syncopal recurrences in patients with prolonged HV implanted with a pacemaker compared with a control group of untreated patients with a negative EPS¹⁸⁸, or with a control group who received an empiric pacemaker.²¹⁷ These results justify an upgrade of the recommendation for EPS-guided therapy (i.e. cardiac pacing) in patients with a positive EPS from class IIa to class I.

Even if the quality of evidence is moderate, there is strong consensus that a positive EPS indicates that the likely mechanism of syncope is paroxysmal AV block.

Conversely, approximately one-third of patients with a negative EPS in whom an ILR was implanted developed intermittent or permanent AV block on follow-up.¹⁸⁷ Thus, EPS has a low negative predictive value.

Mortality is high in patients with syncope and BBB. However, neither syncope nor prolonged H-V interval were associated with a higher risk of death, and pacemaker therapy did not decrease this risk.²¹³

4.2.6.3 Suspected tachycardia

In patients with syncope preceded by a sudden onset of brief palpitations suggesting SVT or VT, an EPS may be indicated to assess the exact mechanism, especially when a curative catheter ablation procedure is considered to be beneficial.

In patients with a previous myocardial infarction and preserved left ventricular ejection fraction (LVEF), induction of sustained monomorphic VT is strongly predictive of the cause of syncope,²¹⁸ whereas the induction of ventricular fibrillation (VF) is considered a non-specific finding.³⁷ The absence of induction of ventricular arrhythmias identifies a group at lower risk of arrhythmic syncope.²¹⁹

The role of EPS and the use of pharmacological challenge by class I antiarrhythmic drugs in patients with syncope and suspected Brugada syndrome is controversial. In a recent meta-analysis,²²⁰ the risk of arrhythmic events was slightly increased in patients with a history of unexplained syncope or a spontaneous type 1 pattern, and who had induction of VT or VF with one or two extra stimuli. However, the absence of induction in such individuals does not necessarily preclude arrhythmia risk, particularly in patients with high-risk features.

[Open in new tab](#)

Electrophysiological study



BBB = bundle branch block; DCM = dilated cardiomyopathy; ECG = electrocardiogram; EPS = electrophysiological study; ESC = European Society of Cardiology; SNRT = sinus node recovery time; SVT = supraventricular tachycardia; VA = ventricular arrhythmia; VF = ventricular fibrillation; VT = ventricular tachycardia.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Electrophysiological study



BBB = bundle branch block; DCM = dilated cardiomyopathy; ECG = electrocardiogram; EPS = electrophysiological study; ESC = European Society of Cardiology; SNRT = sinus node recovery time; SVT = supraventricular tachycardia; VA = ventricular arrhythmia; VF = ventricular fibrillation; VT = ventricular tachycardia.

a

Class of recommendation.

b

Level of evidence.

4.2.7 Endogenous adenosine and other biomarkers

Established cardiac biomarkers such as troponin and B-type natriuretic peptide have been used to distinguish cardiac from non-cardiac syncope and identify structural heart disease. [223a](#) [225](#)

4.2.7.1 Adenosine (triphosphate) test and plasma concentration

The purinergic signalling system, including adenosine and its receptors, has been proposed to be involved in unexplained syncope without prodrome.⁴ A low plasma adenosine level is associated with paroxysmal AV block or CSS, whereas a high level is seen in those with a hypotensive/vasodepressive tendency and VVS. In parallel, the adenosine/adenosine triphosphate (ATP) provocation test has been performed to demonstrate the utility of adenosine sensitivity and paroxysmal cardioinhibitory propensity for the selection of appropriate pacemaker candidates.⁴ The test requires rapid (<20 s) injection of a 20 mg bolus of ATP/adenosine during ECG monitoring. The induction of AV block with ventricular asystole lasting >6 s, or the induction of AV block lasting >10 s, is considered abnormal. ATP testing was positive in most patients with syncope of unknown origin (especially syncope without prodrome and without structural heart disease⁴) but not in controls, suggesting that paroxysmal AV block could be the cause of unexplained syncope. Although cardiac pacing may lead to substantial reduction of syncopal attacks in elderly patients with unexplained syncope and a positive ATP test,²²⁹ previous studies showed no correlation between AV block induced by ATP and ECG findings (documented by ILR) during spontaneous syncope.¹²² Thus, the low predictive value of the test does not support its routine use in selecting patients for cardiac pacing, but rather its positivity suggests that it can be used to confirm the suspicion of asystolic syncope by means of prolonged ECG monitoring. The role of endogenous adenosine release in triggering a special form of asystolic syncope (so-called adenosine-sensitive syncope) remains under investigation.

4.2.7.2 Cardiovascular biomarkers

Some cardiovascular biomarkers are increased in autonomic dysfunction underlying syncope, such as elevated copeptin (vasopressin), endothelin-1, and N-terminal pro-B-type natriuretic peptide in OH,¹¹³ whereas atrial natriuretic peptide may be reduced in POTS.¹¹³ At present, the use of cardiovascular biomarkers in the diagnosis of syncope awaits more evidence and verification in independent cohorts.

4.2.7.3 Immunological biomarkers

Autoantibodies against adrenergic receptors in OH and POTS have been reported, but further studies are needed.²³²

4.2.8 Echocardiography

For patients with suspected heart disease, echocardiography serves to confirm or refute the suspicions in equal proportions and plays an important role in risk stratification.²³⁵ Echocardiography identifies the cause of syncope in very few patients when no more tests are needed (i.e. severe aortic stenosis, obstructive cardiac tumours or thrombi, pericardial tamponade, or aortic dissection).²³⁷ In a literature review, right and left atrial myxoma presented with syncope in <20% of cases.²⁴⁰

4.2.8.1 Exercise stress echocardiography

Upright or semi-supine exercise stress echocardiography to detect provokable left ventricular outflow tract obstruction should be considered in patients with HCM that complain of exertional or postural syncope, particularly when it recurs during similar circumstances (e.g. when rushing upstairs or straining). A gradient of ≥ 50 mmHg is usually considered to be the threshold at which left ventricular outflow tract obstruction becomes haemodynamically important. [245â249](#)

[Open in new tab](#)

Echocardiography



HCM = hypertrophic cardiomyopathy; MRI = magnetic resonance imaging.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Echocardiography



HCM = hypertrophic cardiomyopathy; MRI = magnetic resonance imaging.

a

Class of recommendation.

b

Level of evidence.

4.2.9 Exercise stress testing

Exercise-induced syncope is infrequent, and the literature is limited to case reports. Exercise testing should be performed in patients who have experienced episodes of syncope during or shortly after exertion. Syncope can occur during or immediately after exercise. These two situations should be considered separately. Indeed, syncope occurring during exercise is likely due to cardiac causes (even though some case reports have shown that it might be a manifestation of an exaggerated reflex

vasodilatation), whereas syncope occurring after exercise is almost invariably due to a reflex mechanism.²⁵⁰ Tachycardia-related exercise-induced second- and third-degree AV block has been shown to be located distal to the AV node²⁵³ and predicts progression to permanent AV block.²⁵⁴ A resting ECG frequently shows intraventricular conduction abnormalities,²⁵³ but cases with a normal resting ECG have also been described.²⁵⁶ There are no data supporting an indication for exercise testing in a general population with syncope.

[Open in new tab](#)

Exercise testing



AV = atrioventricular.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Exercise testing



AV = atrioventricular.

a

Class of recommendation.

b

Level of evidence.

4.2.10 Coronary angiography

In patients presenting with syncope and obstructive coronary artery disease, percutaneous coronary intervention is not associated with a significant reduction in readmission for syncope.²⁵⁸ Angiography alone is not diagnostic of the cause of syncope. Therefore, cardiac catheterization techniques should be

carried out in suspected myocardial ischaemia or infarction with the same indications as for patients without syncope.

[Open in new tab](#)

Coronary angiography



a
Class of recommendation.

b
Level of evidence.

[Open in new tab](#)

Coronary angiography



a
Class of recommendation.

b
Level of evidence.

5. Treatment

5.1 General principles of treatment of syncope

The general framework of treatment is based on risk stratification and the identification of specific mechanisms when possible (*Figure 8*).

 General framework of treatment is based on risk stratification and the identification of specific mechanisms when possible. ARVC = arrhythmogenic right ventricular cardiomyopathy; CAD = coronary artery disease; DCM = dilated cardiomyopathy; ECG = electrocardiographic; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; LQTS = long QT syndrome; SCD = sudden cardiac death.

Figure 8

General framework of treatment is based on risk stratification and the identification of specific mechanisms when possible. ARVC = arrhythmogenic right ventricular cardiomyopathy; CAD = coronary artery disease; DCM = dilated cardiomyopathy; ECG = electrocardiographic; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; LQTS = long QT syndrome; SCD = sudden cardiac death.

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The following three general principles should be considered:

- The efficacy of therapy aimed at preventing syncope recurrence is largely determined by the mechanism of syncope rather than its aetiology. Bradycardia is a frequent mechanism of syncope. Cardiac pacing is the most powerful therapy for bradycardia but its efficacy is less if hypotension coexists (see [Table 9](#) and [Supplementary Data Table 9](#)). The treatment of syncope due to a hypotensive reflex or to OH is more challenging because specific therapies are less effective.
- Often, therapy to prevent recurrence differs from that for the underlying disease. The management of patients at high risk of SCD requires careful assessment of the individual patient's risk (see section 5.5).
- Syncopal recurrences often decrease spontaneously after medical assessment, even in the absence of a specific therapy; in general, syncope recurs in <50% of patients within 1â2 years (see [Supplementary Data Table 10](#)). The decrease seems to be more evident when there is a lack of a clear anatomical substrate for syncope, such as in the case of reflex syncope and unexplained syncope. The reason for this decrease is not known. Several potential clinical, statistical, and psychological explanations have been provided and all probably play a role (see [Supplementary Data Table 10](#)). Whatever the reason, the possibility of spontaneous improvement has major practical importance for treatment that can be postponed in low-risk conditions. The consequence of the spontaneous decrease is that any therapy for syncope prevention appears to be more effective than it actually is, which makes the results of observational data on therapy questionable in the absence of a control group.

Table 9

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Expected syncope recurrence rates with a permanent pacemaker in different clinical settings (for more details see [Supplementary Data Table 9](#)).

Clinical setting	Expected 2-year syncope recurrence rate with cardiac pacing
Syncope due to established bradycardia and absence of hypotensive mechanism	High efficacy (â5% recurrence rate)

Syncope due to established bradycardia and associated hypotensive mechanism	Moderate efficacy (5â€â€25% recurrence rate)
Syncope due to suspected bradycardia and associated hypotensive mechanism	Low efficacy (>25% recurrence rate)

Clinical setting	Expected 2-year syncope recurrence rate with cardiac pacing
Syncope due to established bradycardia and absence of hypotensive mechanism	High efficacy (â€â€5% recurrence rate)
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Syncope due to suspected bradycardia and associated hypotensive mechanism	Low efficacy (>25% recurrence rate)

Table 9

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Expected syncope recurrence rates with a permanent pacemaker in different clinical settings (for more details see [Supplementary Data Table 9](#)).

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Syncope due to suspected bradycardia and associated hypotensive mechanism	Low efficacy (>25% recurrence rate)

Clinical setting	Expected 2-year syncope recurrence rate with cardiac pacing
Syncope due to established bradycardia and absence of hypotensive mechanism	High efficacy (â€â€5% recurrence rate)

Syncope due to established bradycardia and associated hypotensive mechanism	Moderate efficacy (5â€25% recurrence rate)
Syncope due to suspected bradycardia and associated hypotensive mechanism	Low efficacy (>25% recurrence rate)

5.2 Treatment of reflex syncope

Despite its benign course, recurrent and unpredictable reflex syncope may be disabling. The cornerstone of management of these patients is non-pharmacological treatment, including education, lifestyle modification, and reassurance regarding the benign nature of the condition.

Additional treatment may be necessary in patients with *severe forms*, as defined in [Web Practical Instructions](#) section 2.3, in particular: when very frequent syncope alters quality of life; when recurrent syncope without, or with a very short, prodrome exposes the patient to a risk of trauma; and when syncope occurs during a high-risk activity (e.g. driving, machine operation, flying, or competitive athletics, etc.). Only 14% of the highly selected population with reflex syncope who are referred to specialized syncope units may need such additional treatment.¹⁸⁶ In general, no therapy is appropriate for every form of reflex syncope. The most important discriminant for the choice of therapy is age. A decision pathway for the selection of a specific therapy according to age, severity of syncope, and clinical forms is summarized in [Figure 9](#).

 Schematic practical decision pathway for the first-line management of reflex syncope (based on patientâ€™s history and tests) according to age, severity of syncope, and clinical forms. Younger patients are those aged <40 years while older patients are >60 years, with an overlap between 40 and 60 years. Severity of reflex syncope is defined in the text. The duration of prodrome is largely subjective and imprecise. A value of â‰¤5 s distinguishes arrhythmic from reflex syncope⁴⁹; in patients without structural heart disease, a duration >10 s can distinguish reflex syncope from cardiac syncope.³⁸ In practice, the prodrome is â€œabsent or very shortâ€œ if it does not allow patients enough time to act, such as to sit or lie down. The heading â€œlow BP phenotypeâ€œ identifies patients with chronic low BP values (in general, systolic around 110 mmHg, who have a clear history of orthostatic intolerance and orthostatic VVS). The group â€œdominant cardioinhibitionâ€œ identifies patients in whom clinical features and results of tests suggest that sudden cardioinhibition is mainly responsible for syncope. One such clue is lack of prodrome, so patients without prodromes may, after analysis, fall into this category.

Figure 9

Schematic practical decision pathway for the first-line management of reflex syncope (based on patientâ€™s history and tests) according to age, severity of syncope, and clinical forms. *Younger patients* are those aged <40 years while *older patients* are >60 years, with an overlap between 40 and 60 years. *Severity* of reflex syncope is defined in the text. The *duration* of prodrome is largely

subjective and imprecise. A value of ≤ 5 s distinguishes arrhythmic from reflex syncope⁴⁹; in patients without structural heart disease, a duration >10 s can distinguish reflex syncope from cardiac syncope.³⁸ In practice, the prodrome is $\hat{\alpha}$ absent or very short $\hat{\alpha}$ if it does not allow patients enough time to act, such as to sit or lie down. The heading $\hat{\alpha}$ low BP phenotyp $\hat{\alpha}$ identifies patients with chronic low BP values (in general, systolic around 110 mmHg, who have a clear history of orthostatic intolerance and orthostatic VVS). The group $\hat{\alpha}$ dominant cardioinhibition $\hat{\alpha}$ identifies patients in whom clinical features and results of tests suggest that sudden cardioinhibition is mainly responsible for syncope. One such clue is lack of prodrome, so patients without prodromes may, after analysis, fall into this category.

Remark:

$\hat{\alpha}$ Overlap between subgroups is expected.

$\hat{\alpha}$ In selected cases, pacing may be used in patients aged <40 years. This Task Force cannot give recommendations due to the lack of sufficient evidence from studies.

$\hat{\alpha}$ In selected cases, fludrocortisone may be used in patients aged >60 years. This Task Force cannot give recommendations due to the lack of sufficient evidence from studies.

$\hat{\alpha}$ Midodrine can be used at any age even if existing studies were performed in young patients.

$\hat{\alpha}$ Patients with short or no prodrome should continue investigations to identify the underlying mechanism and guide subsequent therapy.

$\hat{\alpha}$ Sometimes an ILR strategy should also be considered in patients aged <40 years.

BP = blood pressure; ILR = implantable loop recorder; VVS = vasovagal syncope.

^aSpontaneous or provoked by, sequentially, carotid sinus massage, tilt testing, or ILR.

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5.2.1 Education and lifestyle modifications

Education and lifestyle modifications have not been evaluated in randomized studies, but there is a consensus for implementing them as first-line therapy in all cases. These comprise reassurance about the benign nature of the disease, education regarding awareness and the possible avoidance of triggers and situations (e.g. dehydration and/or hot crowded environments), and the early recognition of prodromal symptoms in order to sit or lie down and activate counter-pressure manoeuvres without delay. If possible, triggers should be addressed directly, such as cough suppression in cough syncope, micturition in the sitting position, etc. Increased intake of oral fluids is also advised. Salt supplementation at a dose of 120 mmol/day of sodium chloride has been proposed.²⁵⁹ In general, $>50\%$ of patients with recurrent syncopal episodes in the 1 or 2 years before evaluation do not have syncopal recurrences in the following 1 or 2 years and, in those with recurrences, the burden of

syncope decreases by >70% compared with the preceding period. The effect of education and reassurance is probably the most likely reason for the decrease in syncope (see [Supplementary Data Table 10](#)). An example of a patient instruction sheet can be found in the [Web Practical Instructions](#) section 9.1: European Society of Cardiology information sheet for patients affected by reflex syncope.

Despite the lack of controlled studies, there is strong consensus that education and lifestyle modifications have a high impact in reducing recurrence of syncope.

5.2.2 Discontinuation/reduction of hypotensive therapy

Careful avoidance of agents that lower BP, i.e. any antihypertensive agents, nitrates, diuretics, neuroleptic antidepressants, or dopaminergic drugs, is key in the prevention of recurrence of syncope. In a small randomized trial²⁶⁰ performed in 58 patients (mean age 74 ± 11 years) affected by vasodepressor reflex syncope diagnosed by tilt testing or CSM, who were taking on average 2.5 hypotensive drugs, discontinuation or reduction of the vasoactive therapy caused a reduction of the rate of the primary combined endpoint of syncope, presyncope, and adverse events from 50 to 19% (hazard ratio 0.37) compared with a control group who continued hypotensive therapy during a follow-up of 9 months. In the Systolic Blood Pressure Intervention Trial,²⁶¹ patients at high cardiovascular risk who were already using antihypertensive drugs targeting a systolic BP of 120 mmHg had an approximately two-fold risk of syncope vs. the control group targeting a systolic BP of 140 mmHg. In a short-term randomized trial²⁶² conducted in 32 patients affected by CSS, withdrawal of vasodilator therapy reduced the magnitude of the vasodepressor reflex induced by CSM.

There is moderate evidence that discontinuation/reduction of hypotensive therapy targeting a systolic BP of 140 mmHg should be effective in reducing syncopal recurrences in patients with hypotensive susceptibility. Further research is likely to have an important impact on our confidence in the estimate.

5.2.3 Physical counter-pressure manoeuvres

Isometric muscle contractions increase cardiac output and arterial BP during the phase of impending reflex syncope. Three clinical studies¹¹⁹ , ¹²⁰ , ²⁶³ and one prospective multicentre randomized trial¹²¹ assessed the effectiveness of physical counter-pressure manoeuvres (PCM) of the legs or arms and showed that they allowed the patient to avoid or delay losing consciousness in most cases. In the Physical Counterpressure Manoeuvres Trial (PC-Trial),¹²¹ 223 patients aged 38 ± 15 years with recurrent reflex syncope and recognizable prodromal symptoms were randomized to receive standardized conventional therapy alone or conventional therapy plus training in PCM. Actuarial recurrence-free survival was better in the PCM group (log-rank $P=0.018$), resulting in a relative risk reduction of 39% (95% CI 11–53). No adverse events were reported. A limitation of this treatment is that it cannot be used in patients with short or absent prodrome and that PCM are less effective in patients older than 60 years.²⁶⁴ An instruction sheet on how to perform PCM can be found in the [Web Practical Instructions](#) section 9.2.

There is moderate evidence that PCM is effective in reducing syncopal recurrences in patients <60 years old with long-lasting recognizable prodromal symptoms.

5.2.4 Tilt training

In highly motivated young patients with recurrent vasovagal symptoms triggered by orthostatic stress, the prescription of progressively prolonged periods of enforced upright posture (so-called tilt training) has been proposed to reduce syncope recurrence.²⁶⁵ While some studies suggested modest benefit with outpatient tilt training,²⁶⁶ most controlled trials reported no significant effect.^{268a} Moreover, this treatment is hampered by the low compliance of patients in continuing the training programme for a long period.

There is sufficient evidence from multiple trials that tilt training has little efficacy in reducing recurrence of syncope in young patients with long-lasting recognizable prodromal symptoms. Further research is unlikely to have an important impact on our confidence in the estimate.

5.2.5 Pharmacological therapy

Pharmacological therapy may be considered in patients who have recurrent syncope despite education and lifestyle modifications including training in PCM. Many drugs have been tested in the treatment of reflex syncope, for the most part with disappointing results. While results have been satisfactory in uncontrolled trials or short-term controlled trials, several long-term placebo-controlled prospective trials have not shown a benefit of the active drug over placebo, with some exceptions.

5.2.5.1 Fludrocortisone

Fludrocortisone, by increasing renal sodium reabsorption and expanding plasma volume, may counteract the physiological cascade leading to the orthostatic vasovagal reflex.²⁷³ The mechanism of action can be compared with that of saline infusion, which has also proved effective in acute tilt test studies.²⁷⁴ The Prevention of Syncope Trial (POST) 2²⁷⁵ enrolled 210 young (median age 30 years) patients with low-normal values of arterial BP and without comorbidities, and randomized them to receive fludrocortisone (titrated at a dosage of 0.05a0.2 mg once per day) or placebo. The primary endpoint showed only a marginal non-significant reduction in syncope in the fludrocortisone group compared with the placebo group (hazard ratio 0.69, 95% CI 0.46a1.03; $P=0.069$), which became more significant when the analysis was restricted to patients who achieved 0.2 mg/day dose stabilization at 2 weeks. The clinical benefit of fludrocortisone therapy was modest: at 12 months, 44% of patients in the fludrocortisone arm continued to suffer syncope, a rate only slightly lower than the 60.5% rate observed in the placebo arm. In the meantime, a similar number of patients discontinued fludrocortisone therapy owing to side effects, thus equating the benefit/risk ratio. Fludrocortisone should not be used in patients with hypertension or heart failure. Fludrocortisone was ineffective in a small randomized double-blind trial in children.²⁷⁶

There is moderate evidence that fludrocortisone may be effective in reducing syncopal recurrences in young patients with low-normal values of arterial BP and without comorbidities. Further

research is likely to have an important impact on our confidence in the estimate of effect.

5.2.5.2 Alpha-agonists

As failure to achieve proper vasoconstriction of the peripheral vessels is common in reflex syncope, alpha-agonist vasoconstrictors (etilefrine and midodrine) have been used. Etilefrine has been studied in a large randomized placebo-controlled double-blind trial.²⁷⁷ During follow-up, patients treated twice daily with etilefrine 25 mg or placebo showed no difference in the frequency of syncope or the time to recurrence. Midodrine [usually 2.5aFAESTS10 mg, three times daily (t.i.d)] has proved effective in small studies but none satisfied the criteria of a pivotal clinical trial. A recent systematic review of these trials²⁷⁸ showed that the confidence in estimates was moderate because of imprecision and publication bias. The most frequent side effects that led to discontinuation of midodrine were supine hypertension, pilomotor reactions, and urinary problems (urinary retention, hesitancy, or urgency). The major limitation of midodrine is frequent dosing, which limits long-term compliance. Overall, these data suggest that chronic pharmacological treatment with alpha-agonists alone may be of little use in reflex syncope and that long-term treatment cannot be advised for occasional symptoms.

There are contrasting results from multiple trials that alpha-agonists may be effective in reducing syncopal recurrences in patients with the orthostatic form of VVS. Further research is likely to have an important impact on our confidence in the estimate.

5.2.5.3 Beta-blockers

Beta-blockers have been presumed to lessen the degree of ventricular mechanoreceptor activation owing to their negative inotropic effect in reflex syncope. This theory has not been supported by the outcome of clinical trials. Beta-blockers failed to be effective in VVS in two randomized double-blind controlled trials.²⁷⁹ Â · Â ²⁸⁰ A rationale for the use of beta-blockers in other forms of neutrally-mediated syncope is lacking. It should be emphasized that beta-blockers may enhance bradycardia in CSS.

There is sufficient evidence from multiple trials that beta-blockers are not appropriate in reducing syncopal recurrences. Desirable and undesirable effects are closely balanced.

5.2.5.4 Other drugs

Paroxetine, a selective serotonin reuptake inhibitor, was effective in one placebo-controlled trial, which included highly symptomatic patients from one institution.²⁸¹ This finding has not been confirmed in other studies and has no experimental support. Conversely, human studies with different subtypes of serotonin-receptor antagonists demonstrated a decreased tolerance to tilt.¹ Â · Â ²⁸² In a small randomized trial, benzodiazepine was as effective as metoprolol.²⁸³ A somatostatin analogue (octreotide)²⁸⁴ was used in a few patients affected by orthostatic intolerance and its effect cannot be properly evaluated.

5.2.5.5 Emerging new therapies in specific subgroups

Low-adenosine phenotype. In a series of case reports, theophylline appeared effective in patients with recurrent sudden onset (pre)syncope who presented with the common biological characteristic of low circulating adenosine levels.²⁸⁵ Theophylline is a non-selective adenosine receptor antagonist that is potentially effective when adenosine is suspected to be involved in the mechanism of syncope. An intra-patient comparison between a period with and a period without theophylline therapy, with the support of prolonged ECG monitoring, showed that symptoms disappeared and the number of prolonged asystolic pauses was impressively reduced from a median of 1.11 per month during 13 months of no treatment to 0 per month during 20 months of theophylline treatment.

Low-norepinephrine phenotype. A mismatch between sympathetic nerve activity and norepinephrine spillover is present in patients with orthostatic VVS.²⁸⁷ Norepinephrine transport inhibitors (reboxetine and sibutramine) lead to a selective increase in sympathetic tone during stress by inhibiting the reuptake of norepinephrine in sympathetic neuronal synapses. In double-blind, randomized, crossover fashion, reboxetine and sibutramine block or attenuate the vasovagal reflex during tilt testing.²⁸⁸ In an open-label prospective clinical study in seven very symptomatic patients who had not responded to any previous treatment, sibutramine achieved 94% suppression of syncopal episodes at 6 months.²⁸⁹

Ganglionic plexus ablation. Radiofrequency ablation of vagal ganglia located close to the sinus node and AV node was reported to abolish the vagal efferent output during VVS in some observational studies and case reports.²⁹⁰ However, owing to a weak rationale, small populations, weak documentation of follow-up results, procedural risks, and lack of control groups, the current evidence is insufficient to confirm the efficacy of vagal ganglia ablation.

5.2.6 Cardiac pacing

Permanent pacemaker therapy may be effective if asystole is a dominant feature of reflex syncope. Establishing a relationship between symptoms and bradycardia should be the goal of the clinical evaluation of patients with syncope and a normal baseline ECG. The efficacy of pacing depends on the clinical setting. A comparative table of results in different settings is reported in the [Supplementary Data Table 9](#). [Figure 10](#) summarizes the recommended indication for pacing.

 Summary of indications for pacing in patients with reflex syncope. CI-CSS = cardioinhibitory carotid sinus syndrome.

Figure 10

Summary of indications for pacing in patients with reflex syncope. CI-CSS = cardioinhibitory carotid sinus syndrome.

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5.2.6.1 Evidence from trials in suspected or certain reflex syncope and electrocardiogram-documented asystole

In two observational studies, cardiac pacing reduced syncope burden in patients with documented asystolic syncope by 92¹⁸⁴ and 83%,²⁰⁰ but did not prevent all syncopal events. In the randomized

double-blind Third International Study on Syncope of Uncertain Etiology (ISSUE)-3 trial,¹⁸⁵ 77 patients who had documentation, by means of ILR, of syncope with ≥ 3 -s asystole or ≥ 6 -s asystole without syncope were randomly assigned to receive either dual-chamber pacing with rate drop response or sensing only. During follow-up, the 2-year estimated rate of syncope recurrence was 57% with pacemaker off and 25% with pacemaker on (log-rank $P=0.039$). The risk of recurrence was reduced by 57%. In the ILR subgroup of the multicentre Syncope Unit Project (SUP) 2 study,²⁹² the estimated rates of syncope recurrence with pacing were 11% at 1 year, 24% at 2 years, and 24% at 3 years, and were significantly lower than the corresponding rates observed in untreated control patients. The above evidence supports a class IIa recommendation

There is sufficient evidence that dual-chamber cardiac pacing should be considered to reduce recurrence of syncope when the correlation between symptoms and ECG is established in patients ≥ 40 years of age with the clinical features of those in the ISSUE studies.

5.2.6.2 Evidence from trials in patients with carotid sinus syndrome

The evidence supporting the benefit of cardiac pacing in patients affected by cardioinhibitory CSS is limited to a few small controlled trials and retrospective observational studies. In a review²⁹³ including 12 studies for a total of 601 paced and 305 unpaced patients, the syncopal recurrence rate during follow-up ranged from 0% to 20% with pacing, whereas the recurrence of syncope was always higher in untreated patients, who showed a rate between 20% to 60%. In a meta-analysis of the three studies²⁹³ with a control group of untreated patients, syncope recurred in 9% of 85 paced patients and in 38% of 91 controls (relative risk 0.24, 95% CI 0.12 to 0.48). In a single-centre registry of 169 consecutive patients treated with pacemakers, the actuarial estimate of syncopal recurrence was 7% at 1 year, 16% at 3 years, and 20% at 5 years.⁹⁰ In the CSS subgroup of the multicentre SUP 2 study,²⁹² the estimated syncope recurrence rates with pacing were 9% at 1 year, 18% at 2 years, and 20% at 3 years, and were significantly lower than the corresponding rates observed in untreated controls, which were 21%, 33%, and 43%, respectively. Given the similar outcome of patients with reflex spontaneous asystolic pauses and those with CSS, this Task Force voted to downgrade the recommendation for pacing in patients with CSS from class I (as in the 2013 ESC Pacing Guidelines²⁹⁴) to class IIa.

Despite the lack of large RCTs, there is sufficient evidence that dual-chamber cardiac pacing should be considered to reduce syncopal recurrences in patients affected by dominant cardioinhibitory CSS.

Two variables are well known to hamper the efficacy of pacing therapy in CSS: the mixed forms⁹³ and ⁹⁸ (see also [Web Practical Instructions](#) section 5) and the association with positivity of tilt testing. Patients who have positive tilt tests have a three-fold greater probability of syncope recurrence after dual chamber pacing than those with negative tilt tests²⁹³ and ²⁹⁵; thus, when tilt testing is positive, caution must be recommended over pacemaker implantation.

5.2.6.3 Evidence from trials in patients with tilt-induced vasovagal syncope

Effectiveness of pacing in patients with tilt-induced VVS has been studied in five multicentre RCTs.^{296a}^{FACTS}³⁰⁰ When combining the results of these trials, 318 patients were evaluated; syncope recurred in 21% of the paced patients and in 44% of unpaced patients ($P < 0.001$). A meta-analysis of all studies suggested a non-significant 17% reduction in syncope from the double-blind studies, and an 84% reduction in the studies where the control group did not receive a pacemaker.³⁰¹ In general, pacing was ineffective in trials that enrolled patients without an asystolic tilt response.²⁹⁹ $\hat{\Delta}$ $\hat{\Delta}$ ³⁰⁰ All of these studies have limitations, and a direct comparison is somewhat difficult because of important differences in study design, largely focused on patient selection. Overall, in typical vasovagal populations, pacing seems to have marginal efficacy.

The rationale for the efficacy of cardiac pacing is that the cardioinhibitory reflex is dominant in some patients, as there is no role for pacing in the preventing vasodilatation and hypotension. In a substudy of the ISSUE-3 trial,³⁰² an asystolic response during tilt testing predicted a similar asystolic form during spontaneous ILR-documented syncope, with a positive predictive value of 86%. In the tilt subgroup of the SUP 2 study,²⁹² among 38 patients with dominant cardioinhibitory reflex (with a mean asystolic pause of $22 \hat{\pm} 16$ s) the estimated rates of syncope recurrence with pacing were 3% at 1 year, 17% at 2 years, and 23% at 3 years; these figures were significantly lower than the corresponding rates observed in untreated controls, and were similar to those observed in patients with CSS or with ECG-documented asystole. In a recent multicentre crossover RCT performed in 46 patients aged >40 years, affected by severely recurrent (>5 episodes during life) cardioinhibitory VVS,³⁰³ during 24-month follow-up, syncope recurred in 4 (9%) patients treated with a dual-chamber pacemaker with closed-loop stimulation compared with 21 (46%) patients who had received a sham pacemaker programmed off ($P=0.0001$).

Adding video recording to tilt testing, Saal *et al.* $\hat{\Delta}$ ²⁰⁵ recently showed, in patients with asystole, that asystole occurred 3 s before syncope or later in one-third of patients, in whom cardioinhibition was too late to have primarily caused syncope; in the other two-thirds of asystolic tilt responses, the cause must have been mainly cardioinhibition or a combination of cardioinhibition and vasodepression.

The clinical presentation is probably as important as tilt test positivity when selecting patients who can benefit from cardiac pacing. The SUP 2 study population was characterized by higher mean age, history of recurrent syncope beginning in middle or older age, and frequent injuries, probably due to presentation without warning.²⁹²

Owing to the contrasting results of the randomized trials, the estimated benefit of dual-chamber pacing in cardioinhibitory tilt-positive patients is weak. Divergence of opinion exists among experts. Further research is very likely to have an important impact on recommendations. Conversely, there is strong consensus that pacing cannot be offered to patients with non-cardioinhibitory tilt-positive response, and further tests (e.g. ILR) are warranted to document the mechanism of the spontaneous reflex.

5.2.6.4 Evidence from trials in patients with adenosine-sensitive syncope

Under this term, classified as a non-classical form of reflex syncope in [Table 3](#), different clinical conditions are included, which have a supposed role of adenosine in the genesis of syncope in common.

A new clinical entity, called idiopathic AV block, has recently been described in patients with a long history of syncope and in whom paroxysmal AV block could be recorded at the time of syncope recurrence.⁵ These patients had an otherwise normal heart and no sign of conduction disease on ECG and EPS; they had very low plasma adenosine levels and a high induction rate of transient complete heart block during exogenous injections of adenosine. No syncope recurrence was observed after permanent cardiac pacing over very long periods of follow-up and there was no permanent AV block.

Similarly, the entity of [low-adenosine syncope](#) has recently been described in patients who have an otherwise unexplained syncope with sudden onset without prodrome, a normal heart, and normal ECG.⁴ The clinical, laboratory, and biological features of these patients are similar to those observed in patients affected by idiopathic paroxysmal AV block. Unlike in VVS, tilt testing is usually negative.⁴ No syncope recurrence was observed after permanent cardiac pacing in 10 patients who had ECG documentation of asystolic pause due to sinus arrest or AV block.²⁸⁶

In a small multicentre trial²²⁷ performed in 80 highly selected elderly patients with unexplained unpredictable syncope who had a positive response to intravenous injection of a bolus of 20 mg of ATP, dual-chamber cardiac pacing significantly reduced the 2-year syncope recurrence rate from 69% in the control group to 23% in the active group.

There is weak evidence that dual-chamber cardiac pacing may be useful in reducing recurrences of syncope in patients with the clinical features of adenosine-sensitive syncope. The documentation of possible bradyarrhythmia in spontaneous syncope remains the preferred eligibility criterion for pacing.

5.2.6.5 Choice of pacing mode

In CSS, a few small controlled studies³⁰⁴ and one registry³⁰⁶ showed that dual-chamber pacing is better than the single chamber ventricular mode in counteracting BP fall during CSM and in preventing symptom recurrences. Even if the quality of evidence is weak, dual-chamber pacing is widely preferred in clinical practice.

In patients with VVS, dual-chamber pacing was used mostly with a rate drop response feature that instituted rapid dual-chamber pacing if the device detected a rapid decrease in HR. A comparison between dual-chamber closed-loop stimulation and conventional dual-chamber pacing has been performed by means of a crossover design in two small studies; these studies showed fewer syncope recurrences with closed-loop stimulation, both in the acute setting during repeated tilt testing³⁰⁷ and during 18-month clinical follow-up.³⁰⁸

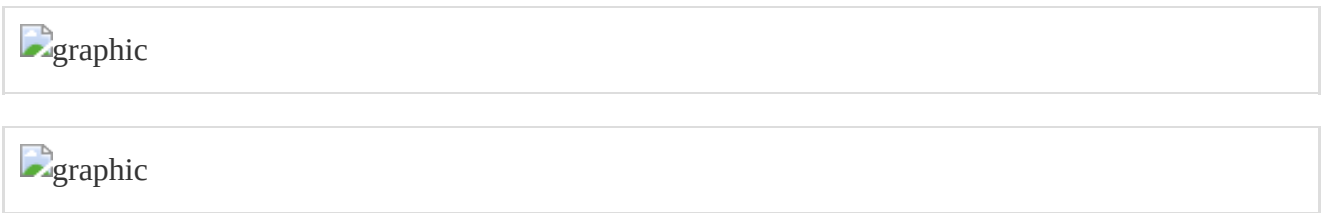
5.2.6.6 Selection of patients for pacing and proposed algorithm

The fact that pacing is effective does not mean that it is always necessary. In patients with reflex syncope, cardiac pacing should be the last choice and should only be considered in highly selected patients, i.e. those ≥ 40 years of age (mostly >60 years), affected by severe forms of reflex syncope with frequent recurrences associated with a high risk of injury, often due to the lack of prodrome.¹⁸⁶ While there is growing scepticism over the diagnostic accuracy of tilt testing for syncope diagnosis, emerging evidence supports the use of tilt testing for the assessment of reflex *hypotensive susceptibility* $\hat{A}$ ¹³², which may be considered to identify patients with an associated hypotensive response who would be less likely to respond to permanent cardiac pacing (see section 4.2.2.2). In a meta-analysis³⁰⁹ of individual patient data from four studies performed in patients with asystolic reflex syncope documented by an ILR, the estimated 3-year recurrence rate of syncope was 2% (95% CI $\hat{A} \pm 4\%$) in tilt-negative patients and 33% (95% CI $\hat{A} \pm 20\%$) in tilt-positive patients; a positive tilt test response was the only significant predictor of syncope recurrence with a hazard ratio of 4.3. Patients with hypotensive susceptibility should need measures directed to counteract hypotensive susceptibility in addition to cardiac pacing, e.g. the discontinuation/reduction of hypotensive drugs and the administration of fludrocortisone or midodrine.

The algorithm shown in *Figure 11* has recently been prospectively validated in a multicentre pragmatic study, which showed a low recurrence rate of syncope with pacing of 9% at 1 year and 15% at 2 years, significantly lower than the 22% and 37%, respectively, observed in unpaced controls.¹⁸⁶

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Treatment of reflex syncope



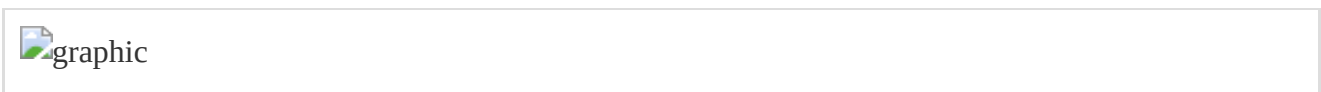
AV = atrioventricular; BP = blood pressure; PCM = physical counter-pressure manoeuvres; VVS = vasovagal syncope.

a
Class of recommendation.

b
Level of evidence.

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Treatment of reflex syncope





AV = atrioventricular; BP = blood pressure; PCM = physical counter-pressure manoeuvres; VVS = vasovagal syncope.

a

Class of recommendation.

b

Level of evidence.

Decision pathway for cardiac pacing in patients with reflex syncope. CI-CSS = cardioinhibitory carotid sinus syndrome; CSM = carotid sinus massage; DDD PM = dual-chamber pacemaker; ILR = implantable loop recorder.

Figure 11

Decision pathway for cardiac pacing in patients with reflex syncope. CI-CSS = cardioinhibitory carotid sinus syndrome; CSM = carotid sinus massage; DDD PM = dual-chamber pacemaker; ILR = implantable loop recorder.

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5.3 Treatment of orthostatic hypotension and orthostatic intolerance syndromes

Current management strategies for OH are summarized in *Figure 12*.

Schematic practical guide for the treatment of orthostatic hypotension.

Figure 12

Schematic practical guide for the treatment of orthostatic hypotension.

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5.3.1 Education and lifestyle measures

Education regarding the nature of the condition in conjunction with the lifestyle advice outlined in section 5.2.1 can markedly improve orthostatic symptoms, even though the rise in BP is relatively small (10–15 mmHg); raising standing BP to just within the autoregulatory zone can make a substantial functional difference. Ambulatory BP recordings may be helpful in identifying abnormal diurnal patterns. These recordings may also help identify supine or nocturnal hypertension in treated patients.

5.3.2 Adequate hydration and salt intake

The expansion of extracellular volume is an important goal. In the absence of hypertension, patients should be instructed to have a sufficient salt and water intake, targeting 2–3 L of fluids per day and 10 g of sodium chloride.³¹⁰ Rapid ingestion of cool water is reported to be effective in combating orthostatic intolerance and postprandial hypotension.³¹¹

5.3.3 Discontinuation/reduction of vasoactive drugs

Several studies that have evaluated the association of vasoactive drugs (i.e. any antihypertensive agents, nitrates, diuretics, neuroleptic antidepressants, or dopaminergic drugs) with OH and falls have yielded contrasting results.³¹² However, intensely prescribed antihypertensive therapy can increase the risk of OH. Intensive antihypertensive treatment can be defined as higher doses of antihypertensive medications, an increased number of antihypertensive drugs, or lowering BP to a target <140/90 mmHg. The total number of BP-lowering medications³¹³ or the use of three or more antihypertensive drugs may be a significant predictor of OH.³¹⁴ Angiotensin-converting enzyme inhibitors, angiotensin receptors blockers, and calcium channel blockers are less likely to be associated with OH compared with beta-blockers and thiazide diuretics.³¹⁵³¹⁸

The principal treatment strategy in drug-induced autonomic failure is eliminating the offending agent. The quality of evidence is moderate. Longer-term future RCTs are likely to have an important impact on determining the net riskâ€”benefit ratio of the withdrawal of culprit medications.

5.3.4 Counter-pressure manoeuvres

PCM such as leg crossing and squatting should be encouraged in patients with warning symptoms who are able to perform them.³¹⁹

5.3.5 Abdominal binders and/or support stockings

Gravitational venous pooling in older patients can be treated with abdominal binders or compression stockings.²³ , ³²⁰ , ³²¹

5.3.6 Head-up tilt sleeping

Sleeping with the head of the bed elevated (>10 degrees) prevents nocturnal polyuria, maintains a more favourable distribution of body fluids, and ameliorates nocturnal hypertension.¹⁰⁴ , ³²² , ³²³

5.3.7 Midodrine

The alpha-agonist midodrine is a useful addition to first-line treatment in patients with chronic autonomic failure. It cannot be regarded as a cure, nor is it helpful in all affected patients, but it is very useful in some. There is no doubt that midodrine increases BP both in the supine and upright posture, and ameliorates the symptoms of OH. Midodrine (2.5â€”10 mg t.i.d) was shown to be effective in three randomized placebo-controlled trials.³²⁴³²⁶

The desirable effects of midodrine outweigh the undesirable effects. The quality of evidence is moderate and further research is likely to have an important impact on the estimate of benefit.

5.3.8 Fludrocortisone

Fludrocortisone (0.1â0.3 mg once daily) is a mineralocorticoid that stimulates renal sodium retention and expands fluid volume.³²⁷ The evidence in favour of fludrocortisone is from two small observational studies (in combination with head-up sleeping) and one double-blind trial in 60 patients; the observational studies showed haemodynamic benefit and, in the trial, treated patients were less symptomatic with higher BP.³²² Â · Â ³²⁷ Â · Â ³²⁸

The desirable effects of fludrocortisone outweigh the undesirable effects. The quality of evidence is moderate and further research is likely to have an important impact on the estimate of benefit.

5.3.9 Additional therapies

Additional and less frequently used treatments, alone or in combination, include desmopressin in patients with nocturnal polyuria, octreotide in postprandial hypotension, erythropoietin in anaemia, pyridostigmine, the use of walking sticks, frequent small meals, and the judicious exercise of leg and abdominal muscles, especially swimming. Their efficacy is less established.¹⁰⁴

5.3.10 Emerging new pharmacological therapy in specific subgroups

Droxidopa, a precursor of norepinephrine, is a centrally and peripherally acting alpha/beta-agonist approved by the US Food and Drug Administration for the treatment of symptomatic neurogenic OH. Droxidopa has recently been investigated for the treatment of neurogenic OH in four short-term RCTs^{329a,332} with a total of 485 patients. They showed a modest increase in standing systolic BP and the symptom benefit of droxidopa over placebo regarding some items of quality of life after 2 weeks of treatment, but its benefit was lost after 8 weeks.³³³ Thus, current evidence is insufficient to confirm the efficacy of droxidopa for long-term use.

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Treatment of orthostatic hypertension



BP = blood pressure; OH = orthostatic hypertension; PCM = physical counter-pressure manoeuvres.

a
Class of recommendation.

b
Level of evidence.

[Open in new tab](#)

Treatment of orthostatic hypertension



BP = blood pressure; OH = orthostatic hypertension; PCM = physical counter-pressure manoeuvres.

a

Class of recommendation.

b

Level of evidence.

5.4 Cardiac arrhythmias as the primary cause

5.4.1 Syncope due to intrinsic sinoatrial or atrioventricular conduction system disease

Current management strategies in patients with syncope due to intrinsic cardiac bradycardia are summarized in *Figure 13*.

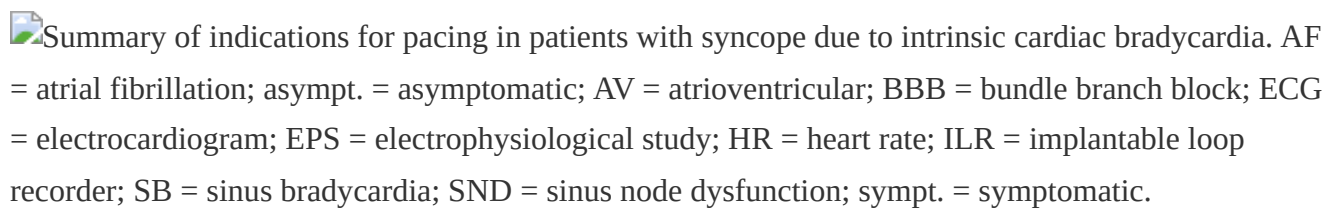
Summary of indications for pacing in patients with syncope due to intrinsic cardiac bradycardia. AF = atrial fibrillation; asympt. = asymptomatic; AV = atrioventricular; BBB = bundle branch block; ECG = electrocardiogram; EPS = electrophysiological study; HR = heart rate; ILR = implantable loop recorder; SB = sinus bradycardia; SND = sinus node dysfunction; sympt. = symptomatic.

Figure 13

Summary of indications for pacing in patients with syncope due to intrinsic cardiac bradycardia. AF = atrial fibrillation; asympt. = asymptomatic; AV = atrioventricular; BBB = bundle branch block; ECG = electrocardiogram; EPS = electrophysiological study; HR = heart rate; ILR = implantable loop recorder; SB = sinus bradycardia; SND = sinus node dysfunction; sympt. = symptomatic.

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5.4.1.1 Sinus node disease

In general, cardiac pacemaker therapy is indicated and has proved effective in intrinsic sinus node disease when intermittent sinus arrest or sinoatrial block has been demonstrated to account for syncope by means of ECG documentation during spontaneous syncope.^{334a}^{FADSTS}³³⁸ A frequent situation is that of patients who have prolonged sinus pause following the termination of tachycardia in bradycardia^{339a}^{FADSTS}³⁴¹ tachycardia syndrome due to the abnormally prolonged time needed for the recovery of automaticity by a diseased sinus node. Permanent pacing does not affect survival.

When the correlation between symptoms and ECG is established, there is general consensus that cardiac pacing is effective and useful for symptom relief.

In the absence of the above situations, despite adequate pacing, syncope recurs in approximately 15a^{FADSTS} 28% of patients at 5 years^{339a}^{FADSTS}³⁴¹ (see [Supplementary Data Table 9](#)). This is due to the

frequent association of a vasodepressor reflex mechanism with sinus node disease. In patients with sinus node disease and syncope, carotid sinus hypersensitivity and a positive response to tilt are present in approximately 50% of patients. Thus, an increased susceptibility to neurally mediated bradycardia/hypotension is often the cause of syncope.¹³⁵ A reflex mechanism of syncope fits well with the unpredictable natural history of syncope recurrence. Physicians should be aware that effectiveness of therapy is not well documented in such cases. From a practical perspective, cardiac pacing may be a reasonable solution in patients affected by sinus node disease, who have had documentation of an asymptomatic ventricular pause >3 s (with exceptions for young trained persons, during sleep, and medicated patients), when a competitive diagnosis, i.e. hypotension, can be ruled out.²⁹⁴ An abnormal SNRT enhances the probability of efficacy of cardiac pacing (see section 4.2.6.1).^{210a, 212}

When the correlation between symptoms and ECG is not established, cardiac pacing may be reasonable in patients with intrinsic sinus node disease, syncope, and documentation of asymptomatic pause(s).

The elimination of drugs that may exacerbate or unmask an underlying susceptibility to bradycardia is an important element in preventing syncope recurrence. Percutaneous cardiac ablative techniques for the control of atrial tachyarrhythmia have become of increasing importance in selected patients with the bradycardia-tachycardia form of sick sinus syndrome, but are infrequently used for the prevention of syncope.

5.4.1.2 Atrioventricular conduction system disease

Cardiac pacing is the treatment of syncope associated with symptomatic AV block (*Figure 13*). Although formal RCTs of pacing in third- or second-degree type 2 AV block have not been performed, some observational studies suggest that pacing is highly effective in preventing syncope recurrences when AV block is documented. Langenfeld *et al.*³⁴¹ observed a decline in the rate of syncope from 44 to 3.4% over 5-year follow-up in 115 patients paced for AV block; the recurrence rate was 7% in the subgroup with syncope before pacemaker implantation. More recently, Sud *et al.*²⁰⁰ reported no syncope recurrence, and Aste *et al.*²⁵⁵ reported a recurrence of 1% at 5 years after pacemaker implantation among 73 patients with documented persistent or intermittent documented AV block (see [Supplementary Data Table 9](#)).

5.4.1.3 Bundle branch block and unexplained syncope

The presence of bifascicular BBB suggests that the cause of syncope may be complete heart block. Nevertheless, less than half of the patients with bifascicular BBB and syncope will have a final diagnosis of AV block, a similar percentage will have a final diagnosis of reflex syncope, and, in approximately 15%, the cause will remain unexplained at the end of a complete workup.³⁴² In addition, among patients receiving an ILR, approximately half remained free of syncope for >2 years after the implantation.¹⁸⁷ Conversely, implantation of a pacemaker

without documentation of AV block (empirical pacing) exposed patients to the risk of recurrence of syncope in about one-quarter of cases during long-term follow-up and was unnecessary in another half.²¹⁷ & & ³⁴⁴ Thus, only one in four pacemakers will finally be appropriate. Finally, pacemaker treatment has not been proven to have a survival benefit. The above considerations justify a class IIb indication in the ESC Guidelines on pacing.²⁹⁴

To overcome the above problems, ESC Guidelines on pacing²⁹⁴ in patients with LVEF >35% recommend a strategy of EPS followed by ILR if the EPS findings are unremarkable. With this strategy, a pacemaker was implanted in approximately half of the patients and these patients had syncope recurrence after pacemaker implantation in 0% of cases¹⁸⁸ & & ²¹⁷. This strategy was safe; however, this Task Force recognizes that in the real world, an empirical pacemaker may be acceptable in selected patients at high risk of traumatic recurrence (e.g. elderly patients with unpredictable syncope) and that an individual risk-benefit evaluation is warranted (Figure 14).

 Therapeutic algorithm for patients presenting with unexplained syncope and bundle branch block. BBB = bundle branch block; CRT-D = cardiac resynchronization therapy defibrillator; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; PM = pacemaker.

Figure 14

Therapeutic algorithm for patients presenting with unexplained syncope and bundle branch block. BBB = bundle branch block; CRT-D = cardiac resynchronization therapy defibrillator; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; PM = pacemaker.

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Even if the quality of evidence is moderate, there is strong consensus that in patients with bifascicular BBB with a positive EPS or documentation of paroxysmal AV block during prolonged ECG monitoring, cardiac pacing is highly effective in preventing syncope recurrence. The evidence of efficacy of empirical pacing strategy is weak and the estimate of benefit is uncertain.

Although syncope is not associated with an increased incidence of sudden death in patients with preserved cardiac function, a high incidence of total deaths (about one-third sudden) was observed in patients with BBB and heart failure, previous myocardial infarction, or low ejection fraction.³⁴⁵ & & ³⁴⁷ Indeed, the high total and sudden mortality seems to be mainly related to underlying structural heart disease and ventricular tachyarrhythmias. In this latter situation, syncope is a risk factor, rather than the cause, of death.²¹⁸ Unfortunately, ventricular programmed stimulation does not seem to identify these patients correctly, and the finding of inducible ventricular arrhythmia (VA) should therefore be interpreted with caution.³⁴⁵ & & ³⁴⁶ Therefore, an implantable cardioverter defibrillator (ICD) or a cardiac resynchronization therapy defibrillator is indicated in patients with

BBB, congestive heart failure, or previous myocardial infarction and depressed systolic function for the prevention of SCD, but may be unable to prevent the recurrence of syncope, which is often due to non-arrhythmic causes such as OH or vasodepressor reflex. The strategy for the management of patients with unexplained syncope and BBB is summarized in *Figure 14*.

5.4.2 Syncope due to intrinsic cardiac tachyarrhythmias

Current management strategies in patients with syncope due to intrinsic cardiac tachyarrhythmia are summarized in *Figure 15*.

Choice of therapy for patients presenting with syncope due to cardiac tachyarrhythmias as the primary cause. AA = antiarrhythmic; ICD = implantable cardioverter defibrillator; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

Figure 15

Choice of therapy for patients presenting with syncope due to cardiac tachyarrhythmias as the primary cause. AA = antiarrhythmic; ICD = implantable cardioverter defibrillator; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

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5.4.2.1 Paroxysmal supraventricular tachycardia

In patients with paroxysmal AV nodal re-entrant tachycardia, AV re-entrant tachycardia, typical atrial flutter, and ectopic tachycardia associated with syncope, catheter ablation is the first-choice treatment. In these patients, the role of drug therapy is limited to being a bridge to ablation or being used when ablation has failed. In patients with syncope associated with atrial fibrillation or atypical left atrial flutter, the decision should be individualized.

5.4.2.2 Paroxysmal ventricular tachycardia

Syncope due to torsade de pointes is not uncommon and is, in its acquired form, the result of drugs that prolong the QT interval. Treatment is the immediate discontinuation of the suspected drug.

Catheter ablation or drug therapy is recommended in patients with syncope due to VT in the presence or absence of structural heart disease in order to prevent syncope recurrence (*Figure 15*). Detailed guidelines regarding antiarrhythmic drug usage in patients with VT can be found in the 2015 ESC Guidelines for VA and the prevention of SCD.⁴⁶

An ICD is indicated in patients with syncope and depressed cardiac function, and VT or VF without correctable cause. Although ICD may not prevent syncope recurrence in these patients,³¹ it is indicated to reduce the risk of SCD (refer to the 2015 ESC Guidelines for VA and the prevention of SCD⁴⁶). An ICD is also indicated in patients with syncope and previous myocardial infarction who have VT induced during EPS³⁴⁶ (see section 4.2.6).

In patients with preserved systolic function, the indication for ICD is weaker because trials have not addressed this specific issue. However, when VT causes syncope, this Task Force believes that an ICD

is warranted if catheter ablation and pharmacological therapy have failed or could not be performed (Figure 15).

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Treatment of syncope due to cardiac arrhythmias



AF = atrial fibrillation; AV = atrioventricular; BBB = bundle branch block; CSM = carotid sinus massage; ECG = electrocardiogram; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Treatment of syncope due to cardiac arrhythmias



AF = atrial fibrillation; AV = atrioventricular; BBB = bundle branch block; CSM = carotid sinus massage; ECG = electrocardiogram; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death; SVT = supraventricular tachycardia; VT = ventricular tachycardia.

a

Class of recommendation.

b

Level of evidence.

5.5 Treatment of syncope secondary to structural cardiac, cardiopulmonary, and great vessel disease

Cardiac syncope is diagnosed when syncope presents in patients with severe aortic stenosis, acute myocardial infarction/ischaemia, HCM, cardiac masses (atrial myxoma, ball thrombus, etc.), pericardial disease/tamponade, congenital anomalies of the coronary arteries, prosthetic valve dysfunction, pulmonary embolus, acute aortic dissection, and pulmonary hypertension (see section 4.1.1). Structural cardiac or cardiopulmonary disease can be present in some patients with syncope, and its incidence increases in older patients. The mere presence of heart disease does not imply that syncope is related to the underlying cardiac disorder. Some of these patients have typical reflex syncope; in others, such as those with inferior myocardial infarction or aortic stenosis, the underlying cardiac disease may play a role in triggering or potentiating a reflex mechanism, and finally the underlying cardiac disease may be the substrate for conduction disturbances, supraventricular arrhythmia, or VA that causes syncope.

Even in the absence of specific trials, there is strong consensus that with syncope secondary to structural cardiac disease, the goal of treatment is not only to prevent syncopal recurrence, but to treat the underlying disease and decrease the risk of death.

5.6 Treatment of unexplained syncope in patients at high risk of sudden cardiac death

The underlying clinical situation is that of a patient being evaluated for ICD implantation because they are affected by syncope(s) supposedly due to transient self-terminating ventricular tachyarrhythmias (fast VT or VF), which have not yet been documented because of their short duration.³⁴⁹ Syncope due to documented VT/VF is outside the scope of this section; please refer to section 5.4.2. General guidance may be sought in the 2015 ESC Guidelines for VA and the prevention of SCD.⁴⁶

5.6.1 Definition

In general, a history of syncope in patients with structural heart disease or inheritable arrhythmia syndromes is associated with a two- to four-fold increased risk of death,³⁴⁸ but varies between specific conditions.³⁵⁴ Moreover, there have been very few studies on ICDs in patients with syncope associated with left ventricular dysfunction,³¹ cardiomyopathy, or inheritable arrhythmia syndromes.³⁵⁷ In these Guidelines, we complement previous ESC Guidelines for VA and the prevention of SCD⁴⁶ by providing a precise definition of unexplained syncope, and making recommendations for its investigation and management in different clinical settings.

- For this section, **unexplained syncope** is defined as syncope that does not meet any class I diagnostic criterion defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a **suspected arrhythmic syncope**.

When the mechanism of syncope is non-arrhythmic, the management of patients at high risk of SCD is the same as for patients without syncope.

5.6.2 Left ventricular systolic dysfunction

The benefit of an ICD to reduce the risk of death is established. Thus, patients with unexplained syncope who have an established ICD indication per current Guidelines⁴⁶ must receive an ICD before, and independently of, the evaluation of the mechanism of syncope, even if the mechanism of syncope is unknown or uncertain at the end of a complete workup. While this strategy may help to prolong life, patients often remain at risk of recurrent syncope, implying a need for precise identification of the mechanism of syncope and specific treatment when possible.

Few data exist concerning the prevalence and the prognostic implications of unexplained syncope in unselected patients with left ventricular dysfunction or non-ischaemic dilated cardiomyopathy with less severe systolic impairment who do not meet the current indication for ICD.³⁵⁸ Data from observational studies in selected cohorts show a high rate of occurrence of ventricular arrhythmias, ICD discharge, and death in patients with a history of unexplained syncope but, owing to a lack of control groups, are unable to show the benefit of an ICD.²⁷ ²⁸ ³⁵⁹ ³⁶⁰ This Task Force believes that an ICD should be considered in patients with unexplained syncope with systolic impairment but without a current indication for ICD to reduce the risk of sudden death.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and left ventricular systolic dysfunction

 graphic

 graphic

ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; LVEF = left ventricular ejection fraction; NYHA = New York Heart Association; SCD = sudden cardiac death.

a
Unexplained syncope is defined as syncope that does not meet a class I diagnostic criterion defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b
Class of recommendation.

c
Level of evidence.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and left ventricular systolic dysfunction



ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; LVEF = left ventricular ejection fraction; NYHA = New York Heart Association; SCD = sudden cardiac death.

a
Unexplained syncope is defined as syncope that does not meet a class I diagnostic criterion defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b
Class of recommendation.

c
Level of evidence.

5.6.3 Hypertrophic cardiomyopathy

Unexplained syncope is an independent predictor for SCD and appropriate ICD discharge. In a systematic review, the average hazard ratio of unexplained syncope (irrespective of definition) was 2.68 (95% CI 0.97 to 4.38).³⁶¹ In the largest multicentre study to date (>3600 patients with HCM), syncope was an independent predictor of the composite of SCD and ICD discharge (hazard ratio 2.05, 95% CI 1.48 to 2.82).³⁵⁰ A prophylactic ICD is appropriate in individuals with other features indicative of a high risk of SCD that are used to estimate the 5-year risk of SCD using the HCM Risk-SCD model²⁴⁵; they include: age, family history of SCD, maximum left ventricular wall thickness, left atrial diameter, and non-sustained VT.

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Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and hypertrophic cardiomyopathy



ESC = European Society of Cardiology; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a

Unexplained syncope is defined as syncope that does not meet the class I diagnostic criterion defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

d

A web-based calculator of the HCM risk score can be found at:

<http://www.doc2do.com/hcm/webHCM.html>. It can also be found in the ESC Pocket Guidelines App found in all app stores.

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Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and hypertrophic cardiomyopathy



ESC = European Society of Cardiology; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a

Unexplained syncope is defined as syncope that does not meet the class I diagnostic criterion defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

d

A web-based calculator of the HCM risk score can be found at:

<http://www.doc2do.com/hcm/webHCM.html>. It can also be found in the ESC Pocket Guidelines App found in all app stores.

5.6.4 Arrhythmogenic right ventricular cardiomyopathy

Although limited and diverse, current data suggest that unexplained syncope is a marker of arrhythmic risk in patients with ARVC.⁴⁶ ³⁵¹ ³⁶² ³⁶³ The decision to implant an ICD should take into account the other known risk factors for arrhythmic events⁴⁶: frequent non-sustained VT, family history of premature sudden death, extensive right ventricular disease, marked QRS prolongation, late gadolinium enhancement on magnetic resonance imaging (MRI) (including left ventricular involvement), left ventricular dysfunction, and VT induction during EPS.⁴⁶

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and arrhythmogenic right ventricular cardiomyopathy



ARVC = arrhythmogenic right ventricular cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a

Unexplained (or uncertain) syncope is defined any syncope that does not meet class I diagnostic criteria defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and arrhythmogenic right ventricular cardiomyopathy



ARVC = arrhythmogenic right ventricular cardiomyopathy; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a

Unexplained (or uncertain) syncope is defined any syncope that does not meet class I diagnostic criteria defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

5.6.5 Patients with inheritable arrhythmogenic disorders

5.6.5.1 Long QT syndrome

Syncopal events in long QT syndrome (LQTS) are associated with an increased risk of subsequent cardiac arrest. The annual rate of SCD in patients with untreated LQTS is around 0.9% overall and 5% for those with syncope.³⁵² , ³⁶⁴ Beta-blocker therapy substantially reduces the risk of syncope and SCD, but presentation with cardiac arrest and recurrent syncope during beta-blocker therapy is associated with the same risk of fatal events as in untreated patients.⁴⁶ For this reason, ICD treatment should be considered in patients with LQTS and recurrent unexplained syncope despite beta-blocker therapy, especially in cases of good treatment compliance, in the absence of precipitating factors, and in LQT2 and LQT3 syndromes. Left cardiac sympathetic denervation should also be considered in this situation, particularly in LQT1.⁴⁶

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Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and long QT syndrome

 graphic

 graphic

ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; LQTS = long QT syndrome; SCD = sudden cardiac death.

a

Unexplained (or uncertain) syncope is defined as any syncope that does not meet class I diagnostic criteria defined in the tables of recommendations in section 4. In the presence of clinical features

described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and long QT syndrome



ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; LQTS = long QT syndrome; SCD = sudden cardiac death.

a

Unexplained (or uncertain) syncope is defined as any syncope that does not meet class I diagnostic criteria defined in the tables of recommendations in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b

Class of recommendation.

c

Level of evidence.

5.6.5.2 Brugada syndrome

A history of syncope may increase the risk of arrhythmic events up to two- to three-fold compared with that in asymptomatic patients. In the largest registry (1029 patients), the incidence of arrhythmic events (sustained VT or VF, appropriate ICD therapy, or sudden death) in patients with Brugada syndrome was 7.7% per year in those with a history of sudden cardiac arrest, 1.9% per year with syncope, and 0.5% per year in asymptomatic patients.^{[353](#)} However, in a second study, the rate of appropriate ICD shocks was similar in asymptomatic patients and in those with syncope, a difference possibly explained by patient selection and a high rate of non-arrhythmic syncope.^{[355](#)}

On balance, this Task Force believes that it is reasonable to consider an ICD in the case of unexplained syncope. New studies^{[356](#)} · ^{[365](#)} published after the 2015 ESC Guidelines for VA and the prevention

of SCD⁴⁶ showed that non-arrhythmic syncope is frequent in Brugada syndrome and appears to be more benign; thus, ICD should be avoided in patients with non-arrhythmic syncope that is established according to the definition reported in this section. ILR is increasingly used in doubtful cases to exclude a VA as the cause of syncope.³⁶⁵ $\hat{A}$, $\hat{A}$ ³⁶⁶

The final decision to implant an ICD in patients with Brugada syndrome and unexplained syncope should also take into account other risk factors for arrhythmic events, including spontaneous type 1 Brugada ECG pattern, family history of sudden death, VF inducibility with one or two ventricular premature beats during EPS, fractionated QRS, early repolarization in the peripheral leads, increased T_{peak} ^{FABTS} T_{end} interval, and long PR interval.²²⁰ $\hat{A}$, $\hat{A}$ ³⁶⁷^{FABTS}³⁷¹ A drug-induced type 1 ECG pattern has a lower risk of sudden death than a spontaneous type 1 response.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and Brugada syndrome

 graphic

 graphic

ECG = electrocardiogram; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a
Unexplained (or uncertain) syncope is defined as any syncope that does not meet the class I diagnostic criteria defined in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b
Class of recommendation.

c
Level of evidence.

[Open in new tab](#)

Implantable cardioverter defibrillator indications in patients with unexplained syncope^a and Brugada syndrome

 graphic



ECG = electrocardiogram; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; SCD = sudden cardiac death.

a
Unexplained (or uncertain) syncope is defined as any syncope that does not meet the class I diagnostic criteria defined in section 4. In the presence of clinical features described in this section, unexplained syncope is considered a risk factor for ventricular tachyarrhythmias.

b
Class of recommendation.

c
Level of evidence.

5.6.5.3 Other forms

Due to a lack of studies examining unexplained syncope in other forms of inheritable arrhythmic diseases such as catecholaminergic polymorphic VT, early repolarization syndrome, and short QT syndrome, this Task Force is unable to give specific recommendations for the investigation and treatment of unexplained syncope. For further information refer to the 2015 ESC Guidelines for VA and the prevention of SCD.⁴⁶

6. Special issues

6.1 Syncope in patients with comorbidity and frailty

The approach to the assessment and management of an older patient with syncope is similar to that of other age groups; however, there are a number of additional features pertinent to age-related comorbidity and frailty that warrant special attention.^{372a}³⁷⁴

6.1.1 Comorbidity and polypharmacy

Comorbidity influences the diagnosis of syncope and management decisions.³³ · ³⁷⁵ Older patients frequently have abnormal findings on more than one investigation and may have more than one possible cause of syncope.³⁷² · ³⁷⁴ · ³⁷⁶ Conversely, coincidental findings of cardiovascular diagnoses such as aortic stenosis or atrial fibrillation³⁷⁷ may not necessarily be the attributable cause of events.^{378a}³⁸⁰

The prescription of polypharmacy, cardiovascular medications, and psychotropic (neuroleptics and antidepressants) and dopaminergic drugs also increases the risk of syncope and falls.^{381a}³⁸⁵ Conversely, the discontinuation or reduction of hypotensive therapy reduces such risk.²⁶⁰ Negative

dromotropic and chronotropic medications should be carefully evaluated in older patients presenting with syncope or falls.

Focal neurological events can occasionally occur due to hypotension and syncope, even in patients without significant carotid artery stenosis (so called *hypotensive TIA*). Although these neurological events occur in only 6% of patients with recurrent syncope, their misdiagnosis is particularly important because they may lead to a lowering of BP with antihypertensive medications (e.g. if focal neurology is mistakenly attributed to vascular pathology rather than hypotension), and to a further increase of the risk of syncope and neurologic events.³⁸⁶

Despite the lack of large controlled trials and an overall modest quality of studies, there is strong consensus that reduction or discontinuation of hypotensive drugs and psychotropic drugs clearly outweighs the undesirable effects (e.g. complications) of high BP. Further research is likely to have an important impact on our confidence in the estimate of effect.

6.1.2 Falls

Syncopal events may be unwitnessed in over half of older patients meaning that collateral histories may not be available, which makes discrimination between falls and syncope challenging.³⁸⁷ If unwitnessed falls are not due to mechanical slips or trips (i.e. are unexplained or non-accidental), it is likely that the patient experienced a syncopal event and displayed lack of awareness for LOC (Figure 16).³⁸⁸ Management of falls in such circumstances is the same as that for syncope.¹⁹¹

Despite the lack of controlled trials and an overall modest quality of studies, there is strong consensus that the management of unexplained falls should be the same as that for unexplained syncope.

 Flow diagram for the identification of unexplained falls.

Figure 16

Flow diagram for the identification of unexplained falls.

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6.1.3 Cognitive assessment and physical performance tests

Age-related memory impairment or more established forms of cognitive impairment are frequently associated with poor recall and therefore the lack of an accurate history of events. In such circumstances, details of prodromal symptoms, whether or not LOC occurred, and symptoms after the event may be unreliable.³⁷³ Cognitive assessment to inform the accuracy of historical data, and general physical assessment to identify comorbid disorders that influence diagnosis and response to treatments (such as Parkinson's disease, gait and balance abnormalities, previous stroke, and polyneuropathies, etc.), are recommended.

Despite the lack of large controlled trials and an overall modest quality of studies, there is strong consensus that the assessment of older patients with syncope or unexplained falls may require cognitive assessment and physical performance tests in addition to syncope evaluation. Further research is likely to have an important impact on our confidence in the estimate of effect.

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Syncope in patients with comorbidity and frailty



BP = blood pressure; CSM = carotid sinus massage; TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Syncope in patients with comorbidity and frailty



BP = blood pressure; CSM = carotid sinus massage; TIA = transient ischaemic attack; TLOC = transient loss of consciousness.

a

Class of recommendation.

b

Level of evidence.

6.2 Syncope in paediatric patients

6.2.1 Diagnostic evaluation

Diagnostic evaluation in paediatric patients is similar to that in adults.

Two specific conditions³⁹⁹ occur in early childhood:

- Infantile reflex syncopal attacks (also called pallid breath-holding spells or reflex anoxic seizures), elicited by a brief unpleasant stimulus, caused by vagally mediated cardiac inhibition.
- Cyanotic breath-holding spells, characterized by stopping breathing during crying, leading to cyanosis and usually TLOC.

Careful taking of personal and family history and a standard ECG are the most important methods of distinguishing benign reflex syncope (also including reflex anoxic seizure or breath-holding spells) from other causes. If the family history is positive, genetic causes of electrical disease of the heart should be considered first. Some children with reflex syncope also have a positive family history.⁴⁰⁰ Tilt testing seems to have high false-negative and false-positive rates and should be used with caution for the primary identification of reflex syncope. Since tilt protocols commonly used in adults may lack specificity in teenagers, in one study, a shorter tilt test duration of 10 min at 60 or 70 degrees was used and showed a specificity of >85%.⁴⁰¹

In young patients, syncope can rarely be the initial manifestation of unusual but life-threatening conditions such as LQTS, KearnsâSayre syndrome (external ophthalmoplegia and progressive heart block), Brugada syndrome, catecholaminergic polymorphic VT, WolffâParkinsonâWhite syndrome, ARVC, HCM, pulmonary arterial hypertension, myocarditis, arrhythmia after repaired congenital heart disease, and anomalous origin of a coronary artery.

Some aspects of the history can suggest a cardiac origin, and should prompt cardiac evaluation.

- Family history: premature SCD at age <40 years and/or familial heart disease.
- Known or suspected heart disease.
- Event triggers: loud noise, fright, and/or extreme emotional stress.
- Syncope during exercise, including swimming.
- Syncope without prodromes, while supine or sleeping, or preceded by chest pain or palpitations.

6.2.2. Therapy

The therapeutic approach is the same as in adults. However, it should be stressed that the effectiveness of pharmacological agents and tilt training for recurrent reflex syncope is undetermined in the absence of well-designed paediatric trials. Furthermore, even in the presence of VVS with prolonged asystole, pacemakers should be avoided due to the relatively transient and benign nature of the syndrome.⁴⁰²

In summary, the key points for the evaluation of syncope in paediatrics are as follows:

- Syncope in childhood is common, the majority being of reflex origin, with only a minority having a potentially life-threatening cause.
- Discriminating benign from serious causes is made primarily by history, physical examination, and ECG results.
- Children with a history suggesting VVS, a normal ECG, and no family history of arrhythmia should not undergo further cardiac investigations.
- The cornerstone of therapy for young patients with reflex syncope includes education and reassurance.

7. Psychogenic transient loss of consciousness and its evaluation

In psychogenic TLOC there is no gross somatic brain dysfunction, but the attacks fulfil the criteria for TLOC (see section 3.1). There are two types: PPS and PNES. In PPS movements are absent, so PPS resembles syncope or longer-lasting LOC, whereas in PNES impressive limb movements mean the attacks resemble epileptic seizures. PPS and PNES differ pathophysiologically from the TLOC forms that they resemble: in PPS, BP and HR are normal or high rather than low, and the EEG is normal instead of showing the slowing or flattening typical of syncope; in contrast to epileptic seizures, the EEG in PNES shows no epileptiform brain activity during an attack.⁹ · ¹¹⁶

The frequency of PPS and PNES probably depends on the setting. The rate of PPS varies from 1% of patients referred to general syncope clinics⁹⁴ to 8% of patients referred to specialist neurological clinics,¹¹⁶ but PPS is probably insufficiently recognized.¹⁵⁴

7.1 Diagnosis

7.1.1 Historical criteria for attacks

The presence of a psychological trauma is not a prerequisite for a diagnosis of conversion. The diagnosis of PPS rests on positive clues taken from the patient's history and from documenting normal EEG results, HR, or BP during an attack. History taking in PPS usually reveals a combination of the following features¹¹⁶ · ¹⁵⁴ · ⁴⁰³:

1. In most cases, the duration of PPS is as short as in syncope, but a much longer duration is a useful diagnostic finding: patients may lie immobile on the floor for 15¹⁵⁴ 30 min.
2. The eyes are usually open in epileptic seizures and syncope but are usually closed in psychogenic TLOC.
3. The attack frequency is high, with several attacks occurring over a week or in a day.
4. There is usually no recognisable trigger, and no sweating, pallor, or nausea beforehand.

5. Injury does not exclude PNES or PPS.

These features should occur together in most attacks. The presence of another pattern of features suggesting a true syncope type, usually VVS, does not argue against a diagnosis of PPS.

7.1.2 Documentation of key features during an attack

The following features are relevant during an attack:

- Video recording or clinical observation, including provocation of an attack during tilt testing. Primary features: sleep-like body position with closed eyes and lack of response to speech or touch, if tested. Secondary features: subtle signs incompatible with LOC such as eyelid flicker, eyeball movements, swallowing, intact muscle tone, normal movements absent in true unconsciousness, and resistance to eye opening.
- BP: normal or elevated during TLOC.
- EEG: normal waking eye-closed EEG pattern, i.e. usually with alpha activity, during TLOC.

The gold standard for PPS is documenting an attack with a home video recorder or with a tilt testing during which BP, HR, and EEG are normal.¹¹⁶ • • ²⁰⁴ • • ⁴⁰⁴ The gold standard for PNES is documenting an attack with video-EEG monitoring.²⁰⁴ • • ⁴⁰⁴

7.1.2.1 Management of psychogenic pseudosyncope

Announcing a psychological diagnosis to patients may be considered difficult, but is necessary for reasons of honesty and as the first step of treatment.⁴⁰⁴ It should be done by the somatic specialist who diagnoses PPS.¹¹⁶ • • ⁴⁰⁴ Important aspects are to assure patients that they are taken seriously and that attacks are as involuntary as syncope or an epileptic seizure. Acceptance of the diagnosis by patients may be critical for therapy. In one observational study,⁴⁰⁵ communicating and explaining the diagnosis resulted in an immediate reduction of attack frequency, with 39% of patients being asymptomatic during a mean follow-up period of 4 years. Some advice on how to inform the patient is provided in [Web Practical Instructions](#) section 10: *European Society of Cardiology information sheet for patients affected by psychogenic pseudosyncope*.

Cognitive behavioural therapy is the usual treatment of PNES and PPS, if attacks remain present after explanation. One pilot randomized treatment trial, conducted in PNES,⁴⁰⁶ showed that psychological therapy provided more attack reduction than no treatment or treatment with sertraline. There are currently no trials on PPS.

[Open in new tab](#)

Diagnosis and management of psychogenic pseudosyncope





EEG = electroencephalogram; PPS = psychogenic pseudosyncope.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Diagnosis and management of psychogenic pseudosyncope



EEG = electroencephalogram; PPS = psychogenic pseudosyncope.

a

Class of recommendation.

b

Level of evidence.

8. Neurological causes and mimics of syncope

This section discusses neurological disorders causing syncope or resembling it, and tests to be performed in patients with syncope.

8.1 Clinical conditions

8.1.1 Autonomic failure

Neurological evaluation should be considered in OH due to autonomic failure. Warning signs are early impotence, disturbed micturition, hyposmia, rapid eye movement, sleep behaviour disorders,^{[408](#)} ,
^{[409](#)} Parkinsonism, ataxia, cognitive impairment, and sensory deficits. A multidisciplinary approach may be required in secondary autonomic failure and in drug-induced OH, depending on the underlying disease.

8.1.2 Epilepsy and ictal asystole

Table [10](#) provides a number of clues that aid the differentiation of syncope from epileptic seizures.^{[9](#)}
^{[50](#)} , ^{[410](#)} , ^{[411](#)} Epilepsy and syncope may evoke one another on rare occasions, resulting

in epileptic seizures triggering syncope as well as syncope triggering an epileptic seizure. The first form concerns *ictal asystole*. Whereas approximately 90% of all epileptic seizures are accompanied by tachycardia, ictal bradycardia and asystole occur in 0.3â0.5% of seizures.⁴¹² Â· Â· ⁴¹³ Bradycardia precedes asystole and AV block may occur, resembling the ECG pattern of reflex syncope.⁴¹² Â· Â· ⁴¹⁴ Epileptic asystole occurs during partial complex seizures, not during generalized seizures. Epileptic asystole occurs in only a fraction of the seizures of one person, and then occurs after a variable interval of 5â100 s from seizure onset.⁴¹⁵ Â· Â· ⁴¹⁶ If asystole lasts for more than about 8 s, syncope ensues.⁴¹⁶ A typical history is for a partial complex seizure to progress as usual for that patient, and then the patient suddenly falls flaccidly, with or without brief myoclonic jerking.⁴¹⁶ Â· Â· ⁴¹⁷ Ictal bradycardia, asystole, and ictal AV block are likely self-terminating,⁴¹² and are due to vagal activation brought about by the seizure. Cessation of cortical activity due to syncopal cerebral hypoperfusion will end the seizure. Therapy requires anti-epileptic drugs and possibly a pacemaker.⁴¹⁸ Ictal asystole is probably not involved in sudden death in epilepsy, as this typically occurs in patients after unwitnessed nocturnal generalized tonicâclonic seizures, i.e. another type of epilepsy.⁴¹⁴ Â· Â· ⁴¹⁹ Note that most cases of sudden cardiac arrest in patients with epilepsy are due to cardiovascular disease and not to ictal asystole.⁴²⁰

Table 10

[Open in new tab](#)

Differentiating syncope from epileptic seizures^{9,50,410,411}

Clinical feature	Syncope	Epileptic seizures
Useful features		
Presence of trigger	Very often	Rare
Nature of trigger	Differs between types: pain, standing, emotions for VVS; specific trigger for situational syncope; standing for OH	Flashing lights is best known; also range of rare triggers
Prodromes	Often presyncope (autonomic activation in reflex syncope, light-headedness in OH, palpitations in cardiac syncope)	Epileptic aura: repetitive, specific for each patient. Includes <i>dÃ©j vu</i> . Rising sensation in the abdomen (epigastric aura) and/or an unusual unpleasant smell
Detailed characteristics of	<10, irregular in amplitude, asynchronous, asymmetrical	20â100, synchronous, symmetrical, hemilateral

myoclonus	Starts after the onset of LOC	- The onset mostly coincides with LOC - Clear long-lasting automatisms as chewing or lip smacking at the mouth
Tongue bite	Rare, tip of tongue	Side of tongue (rarely bilateral)
Duration of restoration of consciousness	10â30 seconds	May be many minutes
Confusion after attack	No understanding of situation for <10 seconds in most syncope, full alertness and awareness afterwards	Memory deficit, i.e. repeated questions without imprinting for many minutes
Features of limited utility		
Incontinence	Not uncommon	Common
Presence of myoclonus (see below for nature of myoclonus)	Very often	â40%, dependent on accuracy of observation
Eyes open during LOC	Frequent	Nearly always
Fatigue and sleep afterwards	Common, particularly in children	Very common
Blue face	Rare	Fairly often
LOC = loss of consciousness; OH = orthostatic hypotension; VVS = vasovagal syncope.		

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	specific trigger for situational syncope; standing for OH	
Prodromes	Often presyncope (autonomic activation in reflex syncope, light-headedness in OH, palpitations in cardiac syncope)	Epileptic aura: repetitive, specific for each patient. Includes déjà vu. Rising sensation in the abdomen (epigastric aura) and/or an unusual unpleasant smell
Detailed characteristics of myoclonus	• <10, irregular in amplitude, asynchronous, asymmetrical • Starts after the onset of LOC	• 20-100, synchronous, symmetrical, hemilateral • The onset mostly coincides with LOC • Clear long-lasting automatisms as chewing or lip smacking at the mouth
Tongue bite	Rare, tip of tongue	Side of tongue (rarely bilateral)
Duration of restoration of consciousness	10-30 seconds	May be many minutes
Confusion after attack	No understanding of situation for <10 seconds in most syncope, full alertness and awareness afterwards	Memory deficit, i.e. repeated questions without imprinting for many minutes
Features of limited utility		
Incontinence	Not uncommon	Common
Presence of myoclonus (see below for nature of myoclonus)	Very often	~40%, dependent on accuracy of observation
Eyes open during LOC	Frequent	Nearly always
Fatigue and sleep afterwards	Common, particularly in children	Very common

Blue face	Rare	Fairly often
LOC = loss of consciousness; OH = orthostatic hypotension; VVS = vasovagal syncope.		

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[Open in new tab](#)

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Confusion after attack	No understanding of situation for <10 seconds in most syncope, full alertness and awareness afterwards	Memory deficit, i.e. repeated questions without imprinting for many minutes

Features of limited utility		
Incontinence	Not uncommon	Common
Presence of myoclonus (see below for nature of myoclonus)	Very often	~460%, dependent on accuracy of observation
Eyes open during LOC	Frequent	Nearly always
Fatigue and sleep afterwards	Common, particularly in children	Very common
Blue face	Rare	Fairly often
LOC = loss of consciousness; OH = orthostatic hypotension; VVS = vasovagal syncope.		

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The second form concerns a syncopal epileptic seizure. Hypoxia can trigger epileptic seizures.²⁰⁸ Such syncopal epileptic seizures have been described in infants with reflex syncope or cyanotic breath-holding spells. A typical syncopal spell suddenly transforms into prolonged clonic movements that last for minutes; note that shorter epileptic seizures may remain unnoticed.

8.1.3 Cerebrovascular disorders

In general, a TIA concerns a focal neurological deficit without LOC, and syncope the opposite. Subclavian steal refers to the rerouting of blood flow to the arm through the vertebral artery due to proximal stenosis or occlusion of the subclavian artery. A TIA may occur when flow through the vertebral artery cannot supply both the arm and part of the brain during forceful use of the arm. Steal most often affects the left side. When detected with ultrasound, steal is asymptomatic in 64% of cases.⁴²² A TIA is likely due to steal only when it is vertebrobasilar (see below) and associated with exercise of one arm. There are no reliable reports of isolated LOC without focal neurological symptoms and signs in subclavian steal.

A TIA related to a carotid artery does not usually cause TLOC. An exception concerns *orthostatic TIAs*, concerning a combination of multiple stenoses of cerebral arteries and OH. This may rarely result in repetitive, orthostatic, short-lasting, and stereotyped TIAs.⁴²³ Â · Â ⁴²⁴

A TIA of the vertebrobasilar system can cause LOC, but there are always focal signs, usually limb weakness, gait and limb ataxia, vertigo, diplopia, nystagmus, dysarthria, and oropharyngeal dysfunction. Fewer than 1% of patients with vertebrobasilar ischaemia present with a single presenting symptom.⁴²⁵

8.1.4 Migraine

Syncope, presumable VVS, and orthostatic intolerance occur more often in patients with migraine, who have a higher lifetime prevalence of syncope and often frequent syncope.⁴²⁶ In migraineurs, syncope and migraine attacks rarely occur simultaneously.

8.1.5 Cataplexy

Cataplexy concerns paresis or paralysis triggered by emotions, usually laughter, but also by a range of other triggers.⁴²⁷ Patients are conscious even when considered unconscious by eyewitnesses, and there is no amnesia. Cataplexy is a key feature of narcolepsy; other cardinal symptoms are excessive daytime sleepiness, sleep-onset paralysis, and hypnagogic hallucinations. Cataplexy may be mistaken for syncope, but also for PPS: a partial awareness of events may be present in PPS, and the falls of cataplexy are partly controlled because paralysis need not be immediately complete.

8.1.6 Drop attacks

The term drop attacks is confusing as it is variably used for MeniÃ“re's disease, atonic epileptic seizures, and unexplained falls.³⁸⁷ A specific condition also labelled drop attacks concerns middle-aged women (rarely men) who suddenly find themselves falling.⁴²⁸ They usually remember hitting the floor and can stand up immediately afterwards.

[Open in new tab](#)

Neurological evaluation





TLOC = transient loss of consciousness.

a

Class of recommendation.

b

Level of evidence.

[Open in new tab](#)

Neurological evaluation



TLOC = transient loss of consciousness.

a

Class of recommendation.

b

Level of evidence.

8.2 Neurological tests

A schematic comprehensive figure of neurological tests used for autonomic failure is shown in [Figure 17](#).

Diagnostic work-up of cardiovascular autonomic failure (adapted from Fanciulli et al. 429). CNS = central nervous system; CRMP-5 = collapsin response mediator protein 5; DAT = dopamine active transporter; HbA1c = haemoglobin A1c; HIV = human immunodeficiency virus; ¹²³I-MIBG = ¹²³I-metaiodobenzylguanidine; MRI = magnetic resonance imaging; PCA-2 = Purkinje cell cytoplasmic autoantibody type 2; SPECT = single-photon emission computed tomography; SS-A = Sjogren's syndrome-associated antigen A; SS-B = Sjogren's syndrome-associated antigen B.

Figure 17

Diagnostic work-up of cardiovascular autonomic failure (adapted from Fanciulli *et al.* 429). CNS = central nervous system; CRMP-5 = collapsin response mediator protein 5; DAT = dopamine active transporter; HbA1c = haemoglobin A1c; HIV = human immunodeficiency virus; ¹²³I-MIBG = ¹²³I-metaiodobenzylguanidine; MRI = magnetic resonance imaging; PCA-2 = Purkinje cell cytoplasmic

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8.2.1 Electroencephalography

The results of interictal EEGs are normal in syncope.⁴¹⁰ An interictal normal EEG cannot rule out epilepsy and the EEG in epilepsy must always be interpreted in a clinical context. An EEG is not recommended when syncope is the most likely cause of TLOC, but it is when epilepsy is the likely cause or when clinical data are equivocal. The EEG is also useful to establish PPS, if recorded during a provoked attack.

8.2.2 Brain computed tomography and magnetic resonance imaging

Computed tomography and MRI in uncomplicated syncope should be avoided. If neurological examination points out Parkinsonism, ataxia, or cognitive impairment, MRI is recommended. In cases of contraindication for MRI, computed tomography is recommended to exclude brain lesions.

8.2.3 Neurovascular studies

No studies suggest that carotid Doppler ultrasonography is valuable in patients with typical syncope.

8.2.4 Blood tests

An acute or subacute onset of multidomain autonomic failure suggests a paraneoplastic or autoimmune cause. Screening for specific paraneoplastic antibodies is recommended: the most common paraneoplastic antibodies are anti-Hu, while others are anti-Purkinje cell cytoplasmic autoantibody type 2 and anti-collapsin response mediator protein 5.⁴³¹ Seropositivity for any of the above-mentioned antibodies may therefore prompt further investigation for occult malignancy (e.g. whole-body fluorodeoxyglucose¹⁸F-positron emission tomography).⁴³²

Seropositivity for antiganglionic acetylcholine receptor antibodies is the serological hallmark of autoimmune autonomic ganglionopathy.⁴³³

[Open in new tab](#)

Neurological tests

 graphic

 graphic

EEG = electroencephalogram; MRI = magnetic resonance imaging.

a

Class of recommendation.

b
Level of evidence.
[Open in new tab](#)
Neurological tests

 graphic
 graphic

EEG = electroencephalogram; MRI = magnetic resonance imaging.

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Class of recommendation.

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Level of evidence.

9. Organizational aspects

9.1 Syncope (transient loss of consciousness) management unit

Since the publication of the 2009 ECS Guidelines, the EHRA Task Force has published a further position statement on the rationale and requirement for syncope units.⁶³ The position paper offers a pragmatic approach to the rationale and requirement for a syncope unit. It is addressed to physicians and others in administration who are interested in establishing a syncope unit in their hospital so that they can meet the standards proposed by the ESC, the EHRA, and the Heart Rhythm Society. The following is the context and evidence for recommendations regarding syncope units (*Table 11*).

Table 11
[Open in new tab](#)

Key components of a syncope unit

- | |
|--|
| <ul style="list-style-type: none">• The syncope unit should take the lead in service delivery for syncope, and in education and training of healthcare professionals who encounter syncope. |
| <ul style="list-style-type: none">• The syncope unit should be led by a clinician with specific knowledge of TLOC and additional necessary team members (i.e. clinical nurse specialist) depending on the local model of service delivery. |

- The syncope unit should provide minimum core treatments for reflex syncope and OH, and treatments or preferential access for cardiac syncope, falls, psychogenic pseudosyncope, and epilepsy.
- Referrals should be directly from family practitioners, EDs, in-hospital and out-hospital services, or self-referral depending on the risk stratification of referrals. Fast-track access, with a separate waiting list and scheduled follow-up visits, should be recommended.
- Syncope units should employ quality indicators, process indicators, and desirable outcome targets.

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ED = emergency department; OH = orthostatic hypotension; TLOC = transient loss of consciousness.

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9.1.1 Definition of a syncope unit

A syncope unit is a facility featuring a standardized approach to the diagnosis and management of TLOC and related symptoms, with dedicated staff and access to appropriate diagnostics and therapies.

9.1.2 Definition of syncope specialist

The syncope specialist is defined as one who has responsibility for the comprehensive management of the patient from risk stratification to diagnosis, therapy, and follow-up, through a standardized protocol. A syncope specialist is a physician who has sufficient knowledge of historical clues and physical findings to recognize all major forms of TLOC, including mimics, as well as syndromes of orthostatic intolerance.

9.1.3 Goal of a syncope unit

Although the benefit of a syncope unit or a syncope specialist in the different healthcare systems has not been exposed to rigorous scientific or economic scrutiny, the consensus is that a dedicated service (a syncope unit) affords better management of TLOC, from risk stratification to diagnosis, therapy, and follow-up, and better education and training of stakeholders. Further research is likely to have an important impact on our confidence in the estimate of effect.

9.1.4 Model of a syncope unit

The syncope unit should provide minimum core treatments for reflex syncope and OH, and treatments or preferential access for cardiac syncope, falls, psychogenic syncope, and epilepsy (Table 12). The tests and assessments available in the syncope unit are detailed in Table 13.

Table 12

[Open in new tab](#)

Structure of the syncope unit

Staffing of a syncope unit is composed of: Â 1. One or more physicians of any specialty who are syncope specialists. Owing to the multidisciplinary nature of TLOC management, each syncope unit should identify specific specialists for the syncope unit and for consultancies. 2. A staff comprising professionals who will advance the care of patients with syncope. These may be physicians, specialized nurses, or others who bring multidisciplinary skills to the facility, coupled with administrative support. The roles played by members of the team may vary according to local circumstances and individual skill. Nurses may be expected to take very important roles including initial assessment, follow-up clinic evaluation, selection of investigations (including tilt testing), and implantation/insertion of ECG loop recorders according to predefined protocols and local regulations (see Table 14). 3. Given that the syncope unit is integrated within a hospital organization, syncope specialists and staff are not necessarily employed full-time, but frequently have other duties depending on the volume of activity in the unit.
Facility, protocol, and equipment Â

1. A syncope unit will deliver most of its care to outpatients in addition to ED and inpatients.
2. The syncope unit should follow an internal protocol, which applies to diagnosis and management and is agreed by stakeholders.
3. An equipped facility must be available.
4. Essential equipment/tests:
 - o 12-lead ECG and 3-lead ECG monitoring
 - o Non-invasive beat-to-beat BP monitor with recording facilities for subsequent analysis
 - o Tilt-table
 - o Holter monitors/external loop recorders
 - o ILRs
 - o Follow-up of ILRs^a
 - o 24-hour BP monitoring
 - o Basic autonomic function tests.
5. Established procedures for:
 - o Echocardiography
 - o EPS
 - o Stress test
 - o Neuroimaging tests.
6. Specialists' consultancies (cardiology, neurology, internal medicine, geriatric, psychology), when needed.

Therapy

Patients with syncope will receive their therapy under the care of the syncope unit unless expertise outside that of the unit is required.

Database management

Â The syncope unit is required to keep medical records that should also include follow-up when appropriate. The database will also offer the possibility of collaborative research with other syncope units.

BP = blood pressure; ECG = electrocardiogram; ED = emergency department; EPS = electrophysiological study; ILR = implantable loop recorder; TLOC = transient loss of consciousness.

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Table 13

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Test and assessments available in a syncope unit

Initial assessment	
History and physical evaluation including 3-min orthostatic BP measurement ^a	
Â 12-lead standard ECG	
Subsequent tests and assessments (only when indicated)	
Blood tests	Electrolytes, haemoglobin, troponin, B-type natriuretic peptide, glucose, D-dimer, haemogas analysis/oxygen saturation
Provocative tests	CSM, tilt testing
Monitoring	External loop recording, implantable loop recording, ambulatory 1â7 days ECG monitoring, 24â48-hour BP monitoring
Autonomic function tests	Standing test, Valsalva manoeuvre, deep-breathing test, cold pressor test, and/or established procedures for access to other autonomic function tests
Cardiac evaluation	Established procedures for access to echocardiogram, stress test, electrophysiological study, coronary angiography
Neurological evaluation	Established procedures for access to neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)

Geriatric evaluation	Established procedures for access to fall risk assessment (cognitive, gait and balance, visual, environmental) and for gait and balance retraining
Psychological or psychiatric evaluation	Established procedures for access to psychological or psychiatric consultancy (mental health problem or psychogenic syncope)

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Provocative tests	CSM, tilt testing
Monitoring	External loop recording, implantable loop recording, ambulatory 14 days ECG monitoring, 24-hour BP monitoring
Autonomic function tests	Standing test, Valsalva manoeuvre, deep-breathing test, cold pressor test, and/or established procedures for access to other autonomic function tests
Cardiac evaluation	Established procedures for access to echocardiogram, stress test, electrophysiological study, coronary angiography
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Psychological or psychiatric evaluation	Established procedures for access to psychological or psychiatric consultancy (mental health problem or psychogenic syncope)

BP = blood pressure; CSM = carotid sinus massage; ECG = electrocardiogram; EEG = electroencephalogram.

a

Postural orthostatic tachycardia may require longer period of standing.

Table 13

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Test and assessments available in a syncope unit

Initial assessment	
History and physical evaluation including 3-min orthostatic BP measurement ^a	
A 12-lead standard ECG	
Subsequent tests and assessments (only when indicated)	
Blood tests	Electrolytes, haemoglobin, troponin, B-type natriuretic peptide, glucose, D-dimer, haemogas analysis/oxygen saturation
Provocative tests	CSM, tilt testing
Monitoring	External loop recording, implantable loop recording, ambulatory 1â7 days ECG monitoring, 24â48-hour BP monitoring
Autonomic function tests	Standing test, Valsalva manoeuvre, deep-breathing test, cold pressor test, and/or established procedures for access to other autonomic function tests
Cardiac evaluation	Established procedures for access to echocardiogram, stress test, electrophysiological study, coronary angiography
Neurological evaluation	Established procedures for access to neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)
Geriatric evaluation	Established procedures for access to fall risk assessment (cognitive, gait and balance, visual, environmental) and for gait and balance retraining
Psychological or psychiatric evaluation	Established procedures for access to psychological or psychiatric consultancy (mental health problem or psychogenic syncope)

Initial assessment	
History and physical evaluation including 3-min orthostatic BP measurement ^a	

12-lead standard ECG	
Subsequent tests and assessments (only when indicated)	
Blood tests	Electrolytes, haemoglobin, troponin, B-type natriuretic peptide, glucose, D-dimer, haemogas analysis/oxygen saturation
Provocative tests	CSM, tilt testing
Monitoring	External loop recording, implantable loop recording, ambulatory 14 days ECG monitoring, 24-hour BP monitoring
Autonomic function tests	Standing test, Valsalva manoeuvre, deep-breathing test, cold pressor test, and/or established procedures for access to other autonomic function tests
Cardiac evaluation	Established procedures for access to echocardiogram, stress test, electrophysiological study, coronary angiography
Neurological evaluation	Established procedures for access to neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)
Geriatric evaluation	Established procedures for access to fall risk assessment (cognitive, gait and balance, visual, environmental) and for gait and balance retraining
Psychological or psychiatric evaluation	Established procedures for access to psychological or psychiatric consultancy (mental health problem or psychogenic syncope)

BP = blood pressure; CSM = carotid sinus massage; ECG = electrocardiogram; EEG = electroencephalogram.

a

Postural orthostatic tachycardia may require longer period of standing.

9.1.5 Access and referrals to a syncope unit

Referral can be direct from family practitioners, EDs, in-hospital and out-hospital services, or self-referral from the patient. Fast-track access with a separate waiting list and scheduled follow-up visits is recommended. In particular, patients at low/intermediate risk admitted to the ED should benefit from such fast-track facilities (so-called protected discharge or advanced access with an appointment for early assessment) to reduce hospitalization rates, directly from the ED or after a short stay in the short observation unit of the ED (see section 4.1.2).

9.1.6 Outcomes and quality indicators

The EHRA Task Force⁶³ has developed the following preliminary quality indicators, based on consensus, as a rough guide for practitioners:

- 1. Absolute rate of undiagnosed TLOC should be reduced by 20%;
- 2. Less than 20% of low-/intermediate-risk TLOC patients should be admitted from the ED;
- 3. The syncope unit should have a 20% reduction in costs relative to usual practice and improved outcomes (i.e. <5% readmissions for syncope and <20% of paced patients with recurrence at 1 year).

9.2 The clinical nurse specialist in the syncope unit

9.2.1 Definition

The syncope unit clinical nurse specialist is defined as an experienced practitioner who has sufficient knowledge of history features and physical findings to recognize all major forms of TLOC, as well as syndromes of orthostatic intolerance. The clinical nurse specialist should work in close collaboration with the syncope specialist. The core competencies of the clinical nurse specialist include a specialized clinical focus, patient advocacy, education and training, auditing, research, and inter- and intradisciplinary consultations.

9.2.2 Role and skills of the clinical nurse specialist

The clinical nurse specialist should be skilled in the performance and interpretation of structured history taking, 12-lead ECG and routine blood test results, tilt testing, active stand tests, autonomic function tests, ECG monitoring (Holter and/or external loop recorder), ABPM, ILR monitoring, and subsequent triaging of patients and monitoring responses to therapy. Other skills will depend on the service model, e.g. pacemaker interrogation. The clinical nurse specialist may have responsibility for follow-up clinics for cardiovascular risk factor management, autonomic function testing and monitoring, management (including education in PCM) of VVS and OH, and follow-up of external and internal loop and Holter monitors and ABPM⁶³ (Table 14).

Table 14

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The role of physicians and staff in performing procedures and tests

Procedure or test	Syncope unit physician	Syncope unit staff	Non-syncope unit personnel
History taking	X		
Structured history taking (e.g. application of software technologies and algorithms)		X	
12-lead ECG		X	

Blood tests		X	
Echocardiogram and imaging			X
CSM	X		
Active standing test		X	
Tilt testing	(X) ^a	X	
Basic autonomic function test		X	
ECG monitoring (Holter, external loop recorder): administration and interpretation	X	X	
ILR	X	(X) ^b	
Remote monitoring		X	
Other cardiac tests (stress test, EPS, angiograms)			X
Neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)			X
Pacemaker and ICD implantation, catheter ablation			X
Patient education, biofeedback training, ^c and instruction sheet on PCM	X	X	
Final report and clinic note	X		
Communication with patients, referring physicians, and stakeholders.	X	X	
Follow-up	X	X	

Procedure or test	Syncope unit physician	Syncope unit staff	Non-syncope unit personnel
History taking	X		

Structured history taking (e.g. application of software technologies and algorithms)		X	
12-lead ECG		X	
Blood tests		X	
Echocardiogram and imaging			X
CSM	X		
Active standing test		X	
Tilt testing	(X) ^a ₋	X	
Basic autonomic function test		X	
ECG monitoring (Holter, external loop recorder): administration and interpretation	X	X	
ILR	X	(X) ^b ₋	
Remote monitoring		X	
Other cardiac tests (stress test, EPS, angiograms)			X
Neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)			X
Pacemaker and ICD implantation, catheter ablation			X
Patient education, biofeedback training, ^c and instruction sheet on PCM	X	X	
Final report and clinic note	X		
Communication with patients, referring physicians, and stakeholders.	X	X	
Follow-up	X	X	

BP = blood pressure; CSM = carotid sinus massage; ECG = electrocardiogram; EEG = electroencephalogram; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; PCM = physical counter-pressure manoeuvres.

a

Physician need not be in the room, but a physician adequately trained in resuscitation needs to be in the vicinity of the test.

b

Current practice limited to a few countries.

c

Biofeedback means that the PCM training session consists of biofeedback training using a continuous BP monitor. Each manoeuvre is demonstrated and explained. The manoeuvres are practised under supervision, with immediate feedback of the recordings to gain optimal performance.

Table 14

[Open in new tab](#)

The role of physicians and staff in performing procedures and tests

Procedure or test	Syncope unit physician	Syncope unit staff	Non-syncope unit personnel
History taking	X		
Structured history taking (e.g. application of software technologies and algorithms)		X	
12-lead ECG		X	
Blood tests		X	
Echocardiogram and imaging			X
CSM	X		
Active standing test		X	
Tilt testing	(X) ^a	X	
Basic autonomic function test		X	
ECG monitoring (Holter, external loop recorder): administration and interpretation	X	X	

ILR	X	(X) ^{b}	
Remote monitoring		X	
Other cardiac tests (stress test, EPS, angiograms)			X
Neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)			X
Pacemaker and ICD implantation, catheter ablation			X
Patient education, biofeedback training, ^{c} and instruction sheet on PCM	X	X	
Final report and clinic note	X		
Communication with patients, referring physicians, and stakeholders.	X	X	
Follow-up	X	X	

Procedure or test	Syncope unit physician	Syncope unit staff	Non-syncope unit personnel
History taking	X		
Structured history taking (e.g. application of software technologies and algorithms)		X	
12-lead ECG		X	
Blood tests		X	
Echocardiogram and imaging			X
CSM	X		
Active standing test		X	
Tilt testing	(X) ^{a}	X	

Basic autonomic function test		X	
ECG monitoring (Holter, external loop recorder): administration and interpretation	X	X	
ILR	X	(X) ^b	
Remote monitoring		X	
Other cardiac tests (stress test, EPS, angiograms)			X
Neurological tests (computed tomography, magnetic resonance imaging, EEG, video-EEG)			X
Pacemaker and ICD implantation, catheter ablation			X
Patient education, biofeedback training, ^c and instruction sheet on PCM	X	X	
Final report and clinic note	X		
Communication with patients, referring physicians, and stakeholders.	X	X	
Follow-up	X	X	

BP = blood pressure; CSM = carotid sinus massage; ECG = electrocardiogram; EEG = electroencephalogram; EPS = electrophysiological study; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; PCM = physical counter-pressure manoeuvres.

a

Physician need not be in the room, but a physician adequately trained in resuscitation needs to be in the vicinity of the test.

b

Current practice limited to a few countries.

c

Biofeedback means that the PCM training session consists of biofeedback training using a continuous BP monitor. Each manoeuvre is demonstrated and explained. The manoeuvres are practised under supervision, with immediate feedback of the recordings to gain optimal performance.

The clinical nurse specialist should be key in developing and delivering communication strategies and processes for the syncope unit for all stakeholdersâpatients and practitionersâand play a pivotal role in education and training together with the syncope specialist. The clinical nurse specialist should be involved in regular auditing and collection of data to inform quality indicators. See the video in [Web Practical Instructions](#) section 11.

Although the skill mix of a clinical nurse specialist has not been exposed to rigorous scientific or economic scrutiny, the consensus is that the clinical nurse specialist should have the necessary skills to deliver assessment and treatment for syncope and TLOC. Further research is required to establish the benefits.

10. Key messages

The ESC Task Force has selected 19 simple rules to guide the diagnosis and management of syncope patients with TLOC according to the 2018 ESC Guidelines on syncope:

Diagnosis: initial evaluation

1. At the initial evaluation answer the following four key questions:
 - o Was the event TLOC?
 - o In cases of TLOC, are they of syncopal or non-syncopal origin?
 - o In cases of suspected syncope, is there a clear aetiological diagnosis?
 - o Is there evidence to suggest a high risk of cardiovascular events or death?
2. At the evaluation of TLOC in the ED, answer the following three key questions:
 - o Is there a serious underlying cause that can be identified?
 - o If the cause is uncertain, what is the risk of a serious outcome?
 - o Should the patient be admitted to hospital?
3. In all patients, perform a complete history taking, physical examination (including standing BP measurement), and standard ECG.
4. Perform immediate ECG monitoring (in bed or telemetry) in high-risk patients when there is a suspicion of arrhythmic syncope.
5. Perform an echocardiogram when there is previous known heart disease, or data suggestive of structural heart disease or syncope secondary to cardiovascular cause.

6. Perform CSM in patients >40 years of age with syncope of unknown origin compatible with a reflex mechanism.
7. Perform tilt testing in cases where there is suspicion of syncope due to reflex or an orthostatic cause.
8. Perform blood tests when clinically indicated, e.g. haematocrit and cell blood count when haemorrhage is suspected, oxygen saturation and blood gas analysis when hypoxic syndromes are suspected, troponin when cardiac ischaemia-related syncope is suspected, and D-dimer when pulmonary embolism is suspected, etc.

Diagnosis: subsequent investigations

- 9. Perform prolonged ECG monitoring (external or implantable) in patients with recurrent severe unexplained syncope who have all of the following three features:
 - Clinical or ECG features suggesting arrhythmic syncope.
 - A high probability of recurrence of syncope in a reasonable time.
 - Who may benefit from a specific therapy if a cause for syncope is found.
- 10. Perform EPS in patients with unexplained syncope and bifascicular BBB (impending high-degree AV block) or suspected tachycardia.
- 11. Perform an exercise stress test in patients who experience syncope during or shortly after exertion.
- 12. Consider basic autonomic function tests (Valsalva manoeuvre and deep-breathing test) and ABPM for the assessment of autonomic function in patients with suspected neurogenic OH.
- 13. Consider video recording (at home or in hospital) of TLOC suspected to be of non-syncopal nature.

Treatment

- 14. To all patients with reflex syncope and OH, explain the diagnosis, reassure, explain the risk of recurrence, and give advice on how to avoid triggers and situations. These measures are the cornerstone of treatment and have a high impact in reducing the recurrence of syncope.
- 15. In patients with *severe forms of reflex syncope*, select one or more of the following additional specific treatments according to the clinical features:
 - Midodrine or fludrocortisone in young patients with low BP phenotype.
 - Counter-pressure manoeuvres (including tilt training if needed) in young patients with prodromes.

- **ILR-guided management strategy** in selected patients without or with short prodromes.
- **Discontinuation/reduction of hypotensive therapy** targeting a systolic BP of 140 mmHg in old hypertensive patients.
- **Pacemaker implantation** in old patients with dominant cardioinhibitory forms.
- 16. In patients with *OH*, select one or more of the following additional specific treatments according to clinical severity:
 - **Education regarding lifestyle manoeuvres.**
 - **Adequate hydration and salt intake.**
 - **Discontinuation/reduction of hypotensive therapy.**
 - **Counter-pressure manoeuvres.**
 - **Abdominal binders and/or support stockings.**
 - **Head-up tilt sleeping.**
 - **Midodrine or fludrocortisone.**
- 17. Ensure that all patients with cardiac syncope receive the specific therapy of the culprit arrhythmia and/or of the underlying disease.
- 18. Balance the benefits and harm of ICD implantation in patients with unexplained syncope at high risk of SCD (e.g. those affected by left ventricle systolic dysfunction, HCM, ARVC, or inheritable arrhythmogenic disorders). In this situation, *unexplained syncope* is defined as syncope that does not meet any class I diagnostic criterion defined in the tables of recommendations of the 2018 ESC Guidelines on syncope and is considered a *suspected arrhythmic syncope*.
- 19. Re-evaluate the diagnostic process and consider alternative therapies if the above rules fail or are not applicable to an individual patient. Bear in mind that Guidelines are only advisory. Even though they are based on the best available scientific evidence, treatment should be tailored to an individual patient's need.

11. Gaps in evidence and areas for future research

Clinicians responsible for managing patients with TLOC must frequently make treatment decisions without adequate evidence or a consensus of expert opinion. The following is a short list of selected, common issues that deserve to be addressed in future clinical research.

Diagnosis: the gap between the best available scientific evidence and the need for the dissemination of these concepts into clinical practice

There is wide variation in the practice of syncope evaluation, and wide variation in the adoption of recommendations from published guidelines. The absence of a systematic approach to TLOC incurs higher health and social care costs, unnecessary hospitalizations and diagnostic procedures, prolongation of hospital stays, lower diagnostic rates, and higher rates of misdiagnoses and symptom recurrence.

Therefore, there is a need for:

- 1) *Large clinical studies that assess the diagnostic yield and compliance of a guideline-based standardized systematic approach* Despite the recommendation from the ESC Guidelines on syncope, syncope units are not widely established in clinical practice. Barriers to establishing a syncope unit include lack of resources, lack of trained dedicated staff, and complex presentations to multiple settings, necessitating involvement from multiple disciplines. The evidence for the usefulness of syncope units is controversial.

Therefore, there is a need for:

- 2) *Large clinical studies that test the superiority of management in a dedicated syncope facility vs. conventional management*

Diagnosis: the need for new diagnostic tests and devices

BP recording is crucial for the majority of clinical TLOC situations and will yield important information for the treatment of syncope. Unfortunately, current long-term BP (or surrogate) recording systems are not optimal for diagnostic use in the syncope evaluation setting.

Therefore, there is a need for:

- 3) *Development and validation of new diagnostic multiparametric devices that can record heart rhythm and BP (and possibly other physiological parameters such as cerebral saturation or EEG) at the time of a syncopal event.*

Treatment: lack of evidence of efficacy of most available therapies

Only a few small RCTs have been conducted on treatment of syncope. In addition, syncopal recurrences are unpredictable and often decrease spontaneously after medical assessment, even in the absence of a specific therapy. The consequence of the spontaneous decrease is that any therapy for syncope prevention appears to be more effective than it actually is, making the results of observational data on therapy questionable in the absence of a control group. No therapy can be effective for all patients. Any therapy should be assessed in homogeneous subgroups.

Therefore, there is strong urgent need for RCTs on the efficacy of:

- 4) *Pharmacological therapies targeted to specific subgroups of reflex syncope.*
- 5) *Pacemaker therapy targeted to specific subgroups of cardioinhibitory reflex syncope.*
- 6) *Pharmacological therapies of OH-mediated syncope.*
- 7) *ICD therapy targeted to specific subgroups of patients with unexplained syncope at risk of SCD.*

Treatment: the need for new therapies

There is a need to move towards personalized medicine. Improving our knowledge of the biochemical mechanisms underlying specific forms of reflex syncope will allow the development of new therapies in such specific settings. For example, a low adenosine phenotype and a low norepinephrine phenotype have been recently identified.

Therefore, there is a need for:

- 8) *Randomized clinical trials on the efficacy of theophylline (and other xantine antagonists) for low adenosine syncope and norepinephrine transport inhibitors for low epinephrine syncope.*

Syncope is a transient phenomenon. The ideal therapy should be one that is administered only when needed.

Therefore, there is a need for:

- 9) *Randomized clinical trials of on-demand administration of specific therapy based on specific sensors similar to adrenalin injectors in asthma or nasal spray for paroxysmal SVT.*

12. What to do and what not to do messages from the Guidelines

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AF = atrial fibrillation; AV = atrioventricular; BBB = bundle branch block; BP = blood pressure; b.p.m. = beats per minute; CSM = carotid sinus massage; CSS = carotid sinus syndrome; ECG = electrocardiogram; ED = emergency department; EPS = electrophysiological study; ESC = European Society of Cardiology; HR = heart rate; ICD = implantable cardioverter defibrillator; ILR = implantable loop recorder; LVEF = left ventricular ejection fraction; NYHA = New York Heart Association; OH = orthostatic hypotension; SCD = sudden cardiac death; SVT = supraventricular tachycardia; VA = ventricular arrhythmia; VT = ventricular tachycardia; VVS = vasovagal syncope.

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13. Supplementary Data and Web Practical Instructions

[Supplementary Data](#) with additional Web Tables complementing the full text, and an additional [Web Practical Instructions](#) document^{PR05CCH} with a glossary containing definitions of syncope and related concepts with tracings, videos, flow charts, and checklists^{PR05CCH} are available on the *European Heart Journal* website and via the ESC Website at www.escardio.org/guidelines.

14. Appendix

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